

FPT SCHOOL OF BUSINESS & TECHNOLOGY

MASTER THESIS ON SOFTWARE ENGINEERING

HYBRID RETRIEVAL AND LEGAL-DOMAIN ADAPTATION FOR VIETNAMESE LEGAL INFORMATION RETRIEVAL

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Acknowledgements and Declaration of Authorship

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Declaration of Authorship

I declare that this thesis is my own work and has been prepared in accordance with accepted standards of academic integrity. All sources and materials used in this research have been appropriately acknowledged and cited.

The work presented in this thesis has not been submitted, either in whole or in part, for any other degree or academic qualification at this or any other institution.

I take full responsibility for the content, accuracy, and academic integrity of this thesis.

Nghiem Phan Thanh Tuan – 2026

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List of Abbreviations

Terminology follows the retrieval literature and the task definitions introduced in Chapters 3–4 [1, 2, 3, 4].

Abbreviation	Meaning	Explanation
ALQAC	Automated Legal Question Answering Competition	A benchmark and competition for evaluating legal question answering and information retrieval systems.
BCA / MPS	Bộ Công an / Ministry of Public Security collection	A legal question–document collection derived from public questions and published answers associated with Vietnam's Ministry of Public Security.
BM25	Best Matching 25 lexical ranking function	A term-based retrieval method that ranks documents according to query-term matches, term frequency, and document length.
BGE-M3	BGE-M3 multilingual embedding model	A multilingual embedding model used in this thesis as a dense retrieval component.
CV	Cross-validation	An evaluation procedure that partitions data into subsets for model selection and performance estimation.
E5	E5 text-embedding model family	A family of dense embedding models; this thesis evaluates the `multilingual-e5-base` variant.
FT / PT	Fine-tuned / Pretrained	Distinguishes a task-adapted model from its original pretrained version.
IR	Information Retrieval	The task of retrieving and ranking documents that are relevant to a user's information need.
MAP	Mean Average Precision	A ranking metric that averages precision across relevant results and then across queries.
MRR	Mean Reciprocal Rank	A ranking metric based on the reciprocal rank of the first relevant result for each query.
NDCG	Normalized Discounted Cumulative Gain	A ranking metric that rewards relevant documents appearing higher in the ranked result list.

Continued on next page

Abbreviation	Meaning	Explanation
OOF	Out-of-fold	Predictions or results produced for samples excluded from the corresponding fitting fold.
RAG	Retrieval-Augmented Generation	An architecture that retrieves external information before or during text generation.
RQ	Research Question	A specific question that the thesis investigates empirically.
SD	Standard Deviation	A statistical measure describing dispersion around the mean.

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Chapter 1

Introduction

1.1 Research Context

Retrieval-Augmented Generation (RAG) is a technique that grounds language model outputs in external documents retrieved from a target corpus [4]. In legal question answering, retrieval precision is critical: an answer, regardless of how fluent, remains unreliable if the cited legal provisions fail to support it. Accordingly, this thesis centers on retrieval effectiveness. Given a Vietnamese legal query and a corpus of legal documents, the goal is to rank provisions such that relevant articles appear at the top of the retrieved results.

Retrieving Vietnamese legal documents poses several distinct challenges. User queries typically rely on everyday language, whereas statutory provisions are framed in formal legal terminology. Furthermore, articles addressing similar domains often dictate divergent conditions, obligations, or procedural requirements; thus, topical or lexical similarity alone does not establish legal relevance. Collection completeness also sets an upper bound on performance, as unindexed provisions cannot be retrieved. Additionally, misalignments between reference annotations and document identifiers can distort benchmark metrics. A rigorous evaluation must therefore account for ranking quality, corpus coverage, and mapping consistency.

To establish a baseline, this thesis employs BM25, which ranks documents via lexical matching a vital capability given that legal queries frequently contain explicit statutory identifiers, document numbers, or standard phrasing [5, 6]. In contrast, dense retrieval leverages embedding similarity to match queries and documents across lexical gaps. We evaluate two representative pretrained models: BGE-M3 and multilingual-E5-base (E5) [2, 3], along with hybrid configurations (BM25+BGE-M3 and BM25+BGE-M3+E5) that combine lexical and dense retrieval scores.

We evaluate these approaches on benchmark Vietnamese legal datasets using established relevance annotations. Specifically, we investigate whether BM25+BGE-M3 improves upon the BM25 baseline, whether integrating E5 yields further gains, and whether fine-tuning BGE-M3 enhances standalone retrieval and sustains improvements within hybrid pipelines. Through these empirical evaluations, this thesis clarifies the effectiveness of hybrid retrieval strategies for Vietnamese legal information retrieval.

1.2 Research Gap and Approach

Prior research provides a basis for the methods evaluated in this thesis. Rosa et al. report that BM25 is a competitive baseline for legal case retrieval [6]. Chen et al. and Wang et al. present BGE-M3 and multilingual E5 and evaluate their embedding models on multilingual benchmarks [2, 3]. Bruch et al. analyze fusion of lexical and semantic retrieval scores, including the effects of normalization and parameter selection [14]. These findings motivate comparisons between lexical retrieval, dense retrieval, and their combinations.

The scope of that evidence also matters. Legal case retrieval differs from retrieving statutory articles for Vietnamese questions. In heterogeneous retrieval evaluation, BEIR identifies limitations in the generalization of dense retrievers across datasets [17]. Zhan et al. show that training with hard negatives can improve dense retrieval on their evaluated benchmarks [16]; their findings concern dense-retriever training rather than the effect of fine-tuning BGE-M3 within the hybrids evaluated here. Therefore, the literature provides a rationale for testing these methods, while their effectiveness on the selected Vietnamese legal collections remains an empirical question.

The research gap addressed here is the evidence needed for two specific decisions: whether adding E5 improves BM25+BGE-M3, and whether fine-tuning BGE-M3 improves both standalone and hybrid retrieval. The thesis tests these questions under defined corpus conditions and separates model selection from evaluation on a separate test set. This is an empirical contribution rather than a new retrieval algorithm. Chapter 2 reviews the supporting literature, and Chapter 4 specifies the comparisons and evaluation procedures.

The approach has three parts. First, the overall retrieval benchmark evaluates ALQAC and Zalo using only the reference article text supplied with each dataset. All retrieval methods within a dataset use the same questions, corpus, and recorded relevance targets.

Second, a Bộ Công an (BCA) dataset is constructed from published public-service questions and legal citations in their answers. The evaluation examines whether the cited provisions are present in the corpus and identifies retrieval failures. The evaluation result is reported separately due to differences in data source from ALQAC and Zalo.

Third, the BGE-M3 embedding model is fine-tuned using legal question article pairs and evaluated in standalone and hybrid retrieval. BM25 and E5 remain unchanged, and fusion weights are held fixed when comparing pretrained and fine-tuned BGE-M3 within each hybrid. The analysis distinguishes development scores from results on test set queries.

1.3 Research Objectives

The main objective is to evaluate hybrid retrieval and fine-tuning of the BGE-M3 embedding model for Vietnamese legal information retrieval. The thesis also constructs a new dataset

BCA for evaluation, the goal of which is to provide a more comprehensive assessment of retrieval performance in the Vietnamese legal domain. The thesis has four specific objectives:

1. Compare BM25, pretrained BGE-M3, pretrained E5, and BM25+BGE-M3 on ALQAC and Zalo datasets. BM25's evaluation scores serve as the baseline.
2. Determine whether adding E5 improves retrieval by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.
3. Evaluate whether fine-tuning BGE-M3 improves the performance in BM25+BGE-M3 and BM25+BGE-M3+E5.
4. Construct and evaluate the BCA collection from published public-service questions and legal citations in their answers. Benchmark retrieval performance, examine the differences between public questions with Zalo and ALQAC.

Retrieval effectiveness is measured primarily by NDCG@10. Hit@10, MRR@10, and MAP@10 provide additional measures of retrieval performance within the top ten results. Comparisons are reported separately for each dataset and corpus condition to establish the scope of any observed improvement.

1.4 Research Contributions

The thesis contribution is an empirical comparison of retrieval methods performance including BM25, BGE-M3, E5, BM25+BGE-M3, and BM25+BGE-M3+E5 on ALQAC, Zalo, and BCA. The evaluation measures retrieval performance on datasets ALQAC, Zalo, and BCA. The results show that hybrid retrieval improves NDCG@10 over BM25 in most evaluated conditions, while adding E5 provides little to no additional benefit (Tables 5.13 and 5.14).

The second contribution is an evaluation of fine-tuning the BGE-M3 embedding model for standalone and hybrid retrieval. The first experiment selects the fine-tuned model based on hybrid retrieval performance on development data. The second experiment selects the model based on standalone dense retrieval performance on development data. The result shows fine-tuning improves metric NDCG@10 but does not produce a meaningful improvement in either hybrid when fusion weights are the same (Tables 5.8 and 5.9).

The third contribution is the construction and evaluation of the BCA retrieval collection from published public-service questions and legal citations in their answers. The analysis examines the availability of cited provisions in the corpus and dependence among questions linked by shared legal references. The analysis also examines the sortfall failures of the BCA collection (Table 5.2).

1.5 Scope of the Thesis

The thesis evaluates the ranking of legal articles for Vietnamese questions on ALQAC, Zalo, and BCA. It compares BM25, BGE-M3, E5, and their hybrid configurations, and examines the effect of fine-tuning BGE-M3. The evaluation covers retrieval performance, corpus coverage, and the mapping of cited provisions to indexed documents. Answer generation, the quality of legal advice, user satisfaction, the currency of legal documents, and deployment performance are outside its scope.

The evaluation uses Vietnamese questions and legal texts. BGE-M3 and E5 are multilingual models, but the experiments do not assess retrieval in other languages or legal systems. Conclusions are limited to the evaluated datasets, corpus conditions, model configurations, and fusion weights. Applying the findings to other collections or downstream RAG systems requires further evaluation.

For BCA dataset the data is collected from publicly available questions and answers on the Ministry of Public Security's website on the date of collection. Many questions are refrencing private legal documents, which is inaccessible to the public. The evaluation does not attempt to access or retrieve any private legal documents, and the analysis is limited to publicly available information.

1.6 Thesis Organization

Chapter 1: Introduction presents Vietnamese legal information retrieval and the research gap addressed by the thesis. It states the research objectives, contributions, and scope.

Chapter 2: Literature Review reviews prior work on legal retrieval, BM25, multilingual embedding models, hybrid retrieval, and fine-tuning. It also discusses retrieval benchmarks and evaluation methods to establish the basis for the thesis's experimental comparisons.

Chapter 3: Theoretical Background explains lexical and dense retrieval, score normalization and fusion, and contrastive training. It defines relevance, evaluation metrics, and the principles used to account for dependence among queries when interpreting results.

Chapter 4: Research Methodology describes dataset and corpus construction, including the BCA collection, and the retrieval pipeline. It specifies fusion weight selection, BGE-M3 fine-tuning, and the procedures for model selection and evaluation.

Chapter 5: Experiments and Evaluation describes the experimental conditions and benchmark evaluation procedure. Its results and analysis cover the BCA collection, BGE-M3 fine-tuning, and the overall comparison of individual and hybrid retrieval methods, including the contribution of E5.

Chapter 6: Conclusion and Future Work summarizes the findings and contributions in relation to the research objectives. It discusses the limitations of the evaluation and

outlines directions for future work.

Chapter 2

Literature Review

This chapter reviews prior work on legal retrieval, multilingual embeddings, and hybrid fusion. It identifies the research gap addressed by this thesis and positions its contributions relative to the literature.

Figure 2.1 shows the position of retrieval in a Retrieval-Augmented Generation (RAG) system. The retrieval component searches the document sources and supplies information to the generator to support answer generation. If the retriever fails to return relevant information, the generator cannot produce a correct answer. Therefore, retrieval performance is a prerequisite for accurate answer generation.

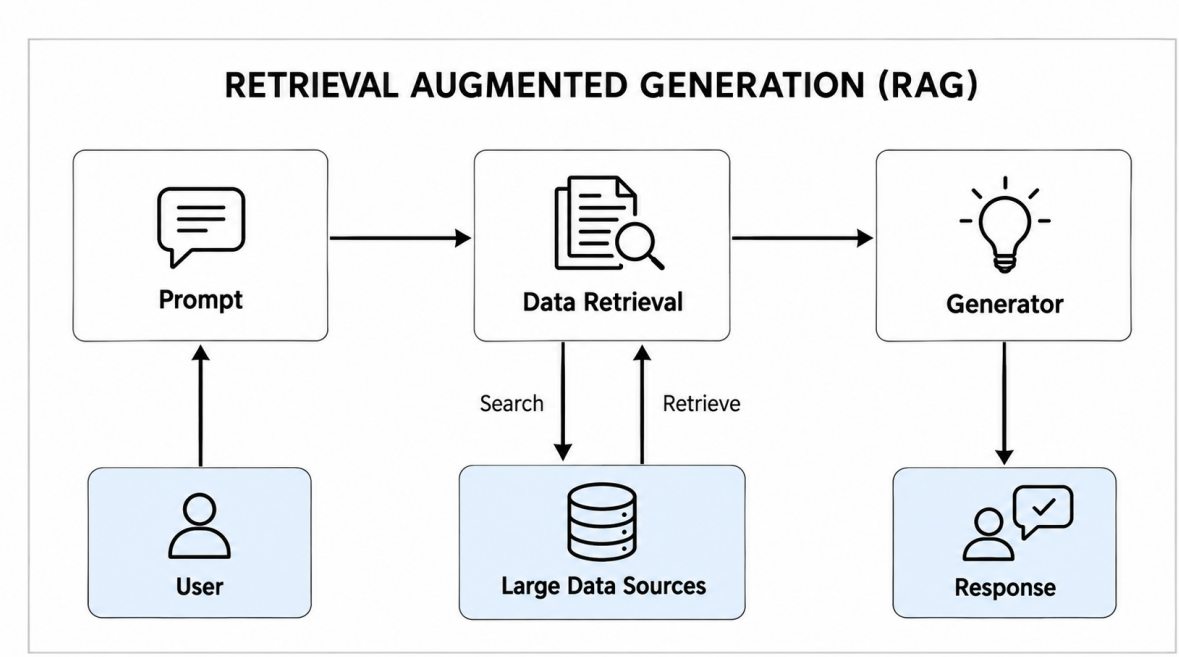


Figure 2.1: Position of retrieval in a Retrieval-Augmented Generation (RAG) system.

2.1 Legal Retrieval and Evidence

Legal information retrieval identifies and ranks legal materials relevant to a query. The retrieval unit may be a judgment, statute, article, or passage. The Competition on Legal Information Extraction and Entailment (COLIEE) separates retrieval tasks from entailment tasks, which assess whether legal texts support a given statement. Its case tasks

use Canadian case law, while its statute tasks use Japanese bar-examination questions [7]. Comparisons with Vietnamese article retrieval must therefore account for differences in the legal system, retrieval unit, relevance annotations, and evaluation metrics.

The Automated Legal Question Answering Competition (ALQAC) 2024 provides manually annotated Vietnamese legal questions and defines separate tasks for legal document retrieval and question answering. In the retrieval task, articles are relevant when they can be used to answer the question [8]. This thesis uses the available relevance annotations to assess how highly the recorded relevant articles are ranked.

Similarity in wording or topic alone does not establish relevance to a legal question. For example, a question about an application deadline may match an article concerning a different stage of the same procedure. In this hypothetical case, the article discusses the same topic but does not provide the required deadline. A relevant provision may also use different terms from those in the question. These possibilities motivate comparing lexical and dense retrieval using the same queries and relevance annotations.

The retrieval unit must also match the relevance annotations. An article may contain rules, exceptions, and references to other provisions, so splitting it into passages changes the context available to the retriever. When the annotations identify articles, retrieved passages must be mapped to their source articles before ranking performance is evaluated. Otherwise, passage retrieval and article retrieval would be assessed using different units.

Text preprocessing is another consideration in Vietnamese retrieval. PhoBERT uses word segmentation before subword tokenization and is evaluated on Vietnamese language-processing tasks [9]. Its preprocessing illustrates a model-specific requirement; its results do not establish that word segmentation improves the retrievers used in this thesis. For lexical retrieval, splitting text at spaces and segmenting it into words produce different vocabularies and term counts. For dense retrieval, preprocessing should follow the embedding model's documented requirements.

In the RAG workflow shown in Figure 2.1, retrieval supplies information to the generator. Lewis et al. combine a language model with retrieved passages for knowledge-intensive tasks [4]. This thesis evaluates the retrieval component by measuring article ranking against the available relevance annotations. Therefore, its results establish retrieval performance within those annotations, while any improvement in the accuracy of generated legal answers requires a separate evaluation.

2.2 Lexical Retrieval and BM25

Lexical retrieval ranks documents by matching query terms with indexed text. BM25 is a probabilistic ranking model that combines term frequency and document frequency with document length normalization [5]. Its scores therefore depend on both the terms in a

document and their distribution in the indexed corpus.

A BM25 submission achieved second place in the legal case retrieval task of COLIEE 2021 [6]. This finding provides empirical support for using BM25 as a baseline in legal retrieval. Since that task concerns case law, its results do not establish BM25's effectiveness for Vietnamese legal article retrieval. The latter requires evaluation on the Vietnamese collections studied here.

This thesis compares dense and hybrid retrieval with BM25 using the same queries, corpus, and relevance annotations within each evaluation condition. BM25's evaluation scores serve as the baseline. Section 3.2 explains the scoring formula, and Section 4.5 specifies the preprocessing and retrieval implementation used in the experiments.

2.3 Dense Retrieval and Multilingual Embeddings

Dense retrieval represents queries and documents as embeddings and ranks documents by vector similarity. Sentence-BERT produces sentence embeddings that can be compared using cosine similarity [10]. Dense Passage Retrieval trains separate query and passage encoders for open-domain question answering [11]. Independent encoding allows document embeddings to be computed before a query is submitted and reused during retrieval.

BGE-M3 supports multilingual retrieval through dense, sparse, and multiple-vector representations [2]. Multilingual E5 provides embedding models in small, base, and large variants, trained using multilingual text pairs and labeled datasets [3]. This thesis evaluates BGE-M3's dense representations and multilingual-E5-base (E5). The results therefore apply to these configurations, rather than to every retrieval mode or model variant described in the source papers.

Single-vector retrieval represents each query and document with one vector. ColBERT instead retains contextualized token embeddings and computes relevance through interactions between query and document tokens after encoding [12]. This approach provides an alternative to single-vector retrieval. ColBERT is included here as related work and is not evaluated in the experiments.

Multilingual-E5-base is included to test whether it complements BGE-M3 in semantic retrieval. The models use different training procedures and embedding methods [2, 3]. The hypothesis is that these differences may produce complementary rankings, allowing E5 to improve retrieval for some questions whose relevant articles are ranked poorly by BGE-M3. Whether this improves hybrid retrieval is tested by comparing BM25+BGE-M3+E5 with BM25+BGE-M3.

The experiments also compare pretrained BGE-M3 and E5 with BM25 to assess standalone retrieval on the Vietnamese legal collections, and BM25+BGE-M3 with BM25 to assess

hybrid retrieval. These comparisons distinguish the effect of combining lexical and dense retrieval from the effect of adding E5 to an existing hybrid.

2.4 Hybrid Retrieval and Score Fusion

Hybrid retrieval combines the outputs of different retrievers to produce a single ranking. Reciprocal Rank Fusion combines rankings using document positions rather than the original retrieval scores [13]. Weighted score fusion instead combines the component scores according to assigned weights. An analysis of lexical and semantic retrieval examines how score normalization and parameter selection affect fusion performance [14]. These findings motivate specifying both the normalization method and the procedure for selecting fusion weights.

The hypothesis is that lexical and dense retrieval may provide complementary rankings. A relevant article ranked poorly by one retriever may be ranked highly by another. Adding E5 to BM25+BGE-M3 tests whether another dense retriever improves a system that already combines lexical and semantic information. Its standalone performance does not establish that additional benefit.

These studies motivate evaluating whether combining lexical and dense retrieval improves Vietnamese legal article ranking, and whether an additional dense retriever provides further benefit. Section 3.4.3 explains the fusion calculation, while Section 4.6 specifies the experimental procedure. Reciprocal Rank Fusion is discussed as related work but is not evaluated in the experiments.

2.5 Fine-Tuning for Legal Retrieval and Hard Negatives

Contrastive training teaches a retrieval model to give relevant passages higher similarity scores than passages labeled as irrelevant [11].

Hard negatives are passages that a retriever ranks highly even though they are labeled as irrelevant. They help train the model to distinguish relevant passages from similar but incorrect matches. Experiments show that mixing hard negatives with randomly selected negatives, or selecting new hard negatives during training, can improve dense retrieval [16].

These findings motivate fine-tuning a dense retriever, and the experiments test whether fine-tuning BGE-M3 can improve retrieval on the Vietnamese legal collections.

2.6 Research Gap and Positioning

BGE-M3 and multilingual E5 provide pretrained embedding models for multilingual retrieval [2, 3]. Research on lexical and semantic score fusion examines the effects of normalization and parameter selection [14], while hard-negative training strategies have improved dense retrieval on benchmark datasets [16]. These studies provide a basis for evaluating multilingual embeddings, score fusion, and embedding model fine-tuning on Vietnamese legal collections.

The research gap addressed by this thesis is the evidence needed to assess these methods under the selected Vietnamese legal datasets. The evaluation compares BM25, BGE-M3, and E5 in standalone and hybrid retrieval, and tests whether fine-tuning BGE-M3 with hard negatives improves retrieval.

The contribution is a controlled empirical comparison of established retrieval methods, together with the construction and evaluation of the BCA collection. The BCA analysis examines the availability of cited provisions and dependence among questions linked by shared legal references. Chapter 4 specifies the datasets and evaluation procedures, and Chapter 5 reports the results within those conditions.

2.7 Chapter Summary

This chapter reviewed legal retrieval, BM25, multilingual embeddings, score fusion, and fine-tuning with hard negatives. The reviewed studies motivate evaluating these methods on Vietnamese legal collections, with separate comparisons for improvements over BM25, the contribution of E5, and the effect of BGE-M3 fine-tuning in standalone and hybrid retrieval. Chapter 3 presents the theoretical background and evaluation metrics used in these comparisons.

Chapter 3

Theoretical Background

This chapter presents the theoretical foundations of retrieval models, score fusion, model training, and evaluation metrics used in the thesis. Chapter 4 describes how these methods are implemented and evaluated.

3.1 Queries, Articles, and Relevance

Information retrieval ranks indexed documents according to their relevance to a query [1]. In this thesis, the query is a Vietnamese legal question, and the retrieval unit is a legal article.

The corpus contains laws (Luật), decrees (Nghị định), and circulars (Thông tư). A document may be organized into parts (phần), chapters (chương), sections (mục), subsections (tiểu mục), articles (điều), clauses (khoản), and points (điểm), although not every document uses all these levels [21]. Each article belongs to one legal document and may contain several clauses and points.

Chunking divides document text into units for indexing and retrieval. The choice of unit can affect retrieval performance [22]. This thesis uses articles as the retrieval units. For documents collected in raw HTML format, the extracted text is separated at article boundaries. Each article record contains its text, document identifier, and article identifier. Clauses and points are kept within their source article. Section 4.2 describes corpus construction and the Zalo reference-text fallback when extraction is unavailable.

Chunking can affect recall because it determines the content available to the retriever before encoding. If a chunk incorrectly combines three articles, one vector represents their combined content. In this thesis, chunking is performed at the article level, so each article receives its own embedding. The retriever can therefore return several articles from the same document, and each article can be evaluated against its own relevance target.

Example of an article-level record. The following example contains two article excerpts from the supplied sample for Decree No. 340/2025/NĐ-CP. The title and article texts are shortened for display, with ellipses marking omitted text. The example illustrates the record format; the complete sample retains each article's clauses and points.

```
{
```

```
  "law_id": "340/2025/NĐ-CP",
  "title": "NGHỊ ĐỊNH SỐ 340/2025/NĐ-CP ...",
  "url": "https://vbpl.vn/van-ban/chi-tiet/nghi-dinh-so-340-2025-nd-cp-quy-dinh-ve-xu
  ↪  -phat-vi-pham-hanh-chinh-trong-linh-vuc-tien-te-va-ngan-hang--187101",
  "articles": [
    {
      "id": "1",
      "text": "Điều 1. Phạm vi điều chỉnh\n1. Nghị định này quy định các hành vi vi
      ↪  phạm hành chính ... trong lĩnh vực tiền tệ và ngân hàng.\n2. Vi phạm hành
      ↪  chính trong lĩnh vực tiền tệ và ngân hàng bao gồm: ..."
    },
    {
      "id": "2",
      "text": "Điều 2. Đối tượng bị xử phạt vi phạm hành chính\n1. Nghị định này áp
      ↪  dụng đối với tổ chức, cá nhân có hành vi vi phạm hành chính trong lĩnh vực
      ↪  tiền tệ và ngân hàng.\n2. Tổ chức quy định tại khoản 1 Điều này gồm: ..."
    }
  ]
}
```


The fields `law_id`, `title`, and `url` describe the legal document. Each entry in `articles` is a separate chunk with its own `id` and `text`. Article 1 and Article 2 therefore receive separate embeddings. Together, `law_id` and the article `id` identify each retrieval unit.

Relevance targets identify the documents or articles recorded as relevant to each query. ALQAC and Zalo use the available dataset annotations. For BCA, the question is used as the query, and citations in its published answer are extracted to form relevance targets. Document names and article numbers are normalized and matched to corpus identifiers. The answer text is used for target extraction and is not supplied to the retriever. Table 4.2 illustrates this process.

An article-level target specifies a document and an article within it. A law-level target specifies only the document. In the dataset, "law-level" also covers decrees and circulars. Duplicate targets are removed, and each target receives evaluation credit at most once. Retrieving several articles from one document therefore does not multiply credit for a single law-level target. Section 3.6 defines the matching rules and evaluation metrics.

Retrieval returns a list of article identifiers ordered by retrieval score. These identifiers link the results to the article text and source document. A RAG generator can use the retrieved text to produce an answer, while this thesis evaluates the article ranking. Corpus coverage measures target availability, and ranking performance measures target positions.

3.2 Lexical Scoring with BM25

BM25 scores articles by matching their indexed terms with those in a query. Term frequency is the number of times a term occurs in an article. Inverse document frequency (IDF) assigns greater weight to terms that occur in fewer documents across the corpus. BM25 combines these quantities with document length normalization [1].

For a query q and an article d, a common BM25 scoring formula sums the contributions of query terms t [5]:

$$\mathrm{BM25}(q,d) = \sum_{t \in q} \mathrm{IDF}(t) \frac{f(t,d)(k_1+1)}{f(t,d) + k_1\left(1 - b + b|d|/\,\mathrm{avgdl}\right)}.$$

The quantities in the formula are:

- $f(t,d)$: the number of occurrences of term t in article d.
- $|d|$: the article length, measured in indexed tokens.
- avgdl: the average article length in the indexed corpus.
- $\mathrm{IDF}(t)$: the inverse document frequency weight of term t.

The parameter k_1 controls term frequency saturation: repeated occurrences of a term contribute progressively less to its score. The parameter b, between zero and one, controls document length normalization. Setting $b = 0$ removes length normalization, while $b = 1$ applies full length normalization [1].

BM25 scores depend on the indexed text. Text extraction can change term counts, article lengths, and corpus statistics, so different corpus representations may produce different scores. A query term absent from an article contributes zero to its BM25 score. Section 4.5 describes the lexical preprocessing used in the experiments.

BM25 relies on matching indexed terms. A relevant article expressed using different terms from the query may therefore receive a low score. The thesis compares BM25 with dense and hybrid retrieval to evaluate whether embedding similarity improves the ranking. BM25 scores determine article order, while evaluation metrics such as NDCG@10 measure ranking quality against the relevance annotations. The baseline is BM25's evaluation performance on the target collections.

3.3 Dense Retrieval and Vector Similarity

An embedding is a numerical vector that represents text. In a bi-encoder architecture, the query and article are encoded separately. Dense retrieval then ranks articles by the similarity

between their embeddings and the query embedding. Sentence-BERT produces sentence embeddings for similarity comparison [10]. Dense Passage Retrieval trains query and passage encoders to retrieve relevant passages [11]. Article embeddings can be computed in advance and reused across queries.

Figure 3.1 illustrates separate query and document encoding, similarity scoring, and ranking.

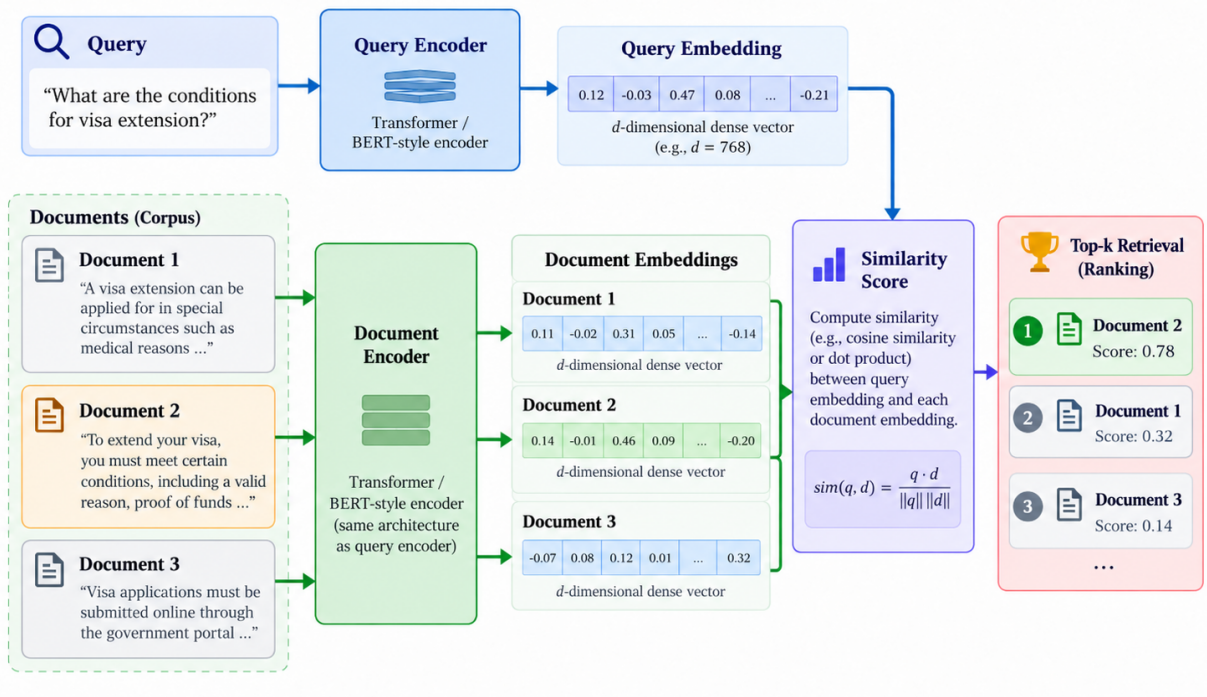


Figure 3.1: Illustrative examples of dense retrieval workflow. The query and document are encoded separately into embeddings, which are then compared using a similarity measure to rank the documents.

For a query q and an article d, u and v are their respective embedding vectors. Two common similarity measures are the dot product and cosine similarity:

$$s_{\text{dot}}(q, d) = u^\top v, \qquad s_{\cos}(q, d) = \frac{u^\top v}{\|u\|_2 \|v\|_2}.$$

The dot product $u^\top v$ sums the products of corresponding vector components. The quantities $\|u\|_2$ and $\|v\|_2$ are the vector lengths. Cosine similarity divides the dot product by these lengths and is defined for nonzero vectors. When both vectors are normalized to length one, the dot product equals cosine similarity.

A pooling operation produces one embedding from the model's token representations. Input longer than the selected token limit is truncated, so the embedding may omit content that remains in the stored article. ColBERT retains multiple token embeddings for matching [12]. This thesis uses single-vector dense retrieval.

3.3.1 BGE-M3

BGE-M3 has three main features:

1. **Multi-Functionality**

   BGE-M3 supports three retrieval methods [24]:

   - **Dense Retrieval:** Uses dense vector representations to capture semantic meaning.
   - **Sparse Retrieval:** Uses sparse representations to match important keywords and lexical features.
   - **Multi-Vector Retrieval:** Uses multiple vectors with late interaction mechanisms, similar to ColBERT, to capture relevance through comparisons between query and document tokens.

2. **Multi-Linguality**

   - Supports more than 100 languages [2].
   - Enables cross-lingual retrieval, allowing a query to retrieve documents written in a different language [2].

3. **Multi-Granularity**

   - Handles text inputs of varying lengths, from short sentences to long documents [24].
   - Supports input sequences of up to 8,192 tokens for encoding short passages and longer documents [24].

This thesis uses BGE-M3's dense representations for retrieval and fine-tuning. The baseline experiments use a maximum input length of 512 tokens (Table 4.3). The hybrid systems combine BM25 scores with BGE-M3 dense similarity scores. BGE-M3's sparse and multiple-vector modes are outside the evaluated configurations.

3.3.2 Multilingual-E5-base

Multilingual E5 models are pretrained on multilingual text pairs and then fine-tuned on labeled datasets [3]. This thesis uses multilingual-E5-base, released as `intfloat/multilingual-e5-base`. The model represents each input text as a vector with 768 components [23]. Its weights remain unchanged throughout the experiments.

E5 is included to test whether combining two dense retrievers improves hybrid retrieval. Its contribution is measured by comparing BM25+BGE-M3+E5 with BM25+BGE-M3 using the same queries, corpus, and relevance annotations within each evaluation condition.

3.4 Score Normalization and combination weights

3.4.1 Hybrid Retrieval

Hybrid retrieval combines scores or rankings from different retrievers. The aim is to improve article ranking by combining lexical matches with embedding similarity. The result depends on how the scores are normalized and combined [14]. This thesis evaluates BM25+BGE-M3 and BM25+BGE-M3+E5.

Scores can be combined for all articles in the corpus or only for candidates selected by the retrievers. In the first approach, each retriever scores every indexed article before the scores are combined. In the second, each retriever first selects its top candidates, and fusion ranks only the articles in those lists. An article missing from every candidate list cannot appear in the final results, regardless of the weights used. This thesis uses the first approach: each retriever scores the full corpus before fusion, as described in Section 4.5.

3.4.2 Score Normalization

The retrievers produce raw scores on different scales:

- **BM25:** The score depends on query term matches, term frequency, inverse document frequency, and article length. It has no fixed numerical range and is not normalized to $[0, 1]$. Its values depend on the query, corpus, and inverse document frequency convention [5].
- **BGE-M3:** The pipeline normalizes query and article embeddings to length one and computes their dot product. For nonzero vectors, this equals cosine similarity, whose theoretical range is $[-1, 1]$.
- **Multilingual-E5-base:** The pipeline uses the same normalized-vector dot product, so its theoretical score range is also $[-1, 1]$. Its scores are computed from E5 embeddings rather than BGE-M3 embeddings.

Section 4.5 specifies the score calculation used in the experiments. Although BGE-M3 and E5 share the same theoretical range, their scores can have different distributions across articles. A raw score from one retriever is therefore not directly interchangeable with a score from another.

Vector normalization and score normalization serve different purposes. Vector normalization gives each embedding a length of one. Min-Max score normalization rescales each retriever's scores across the corpus for a particular query before weighted fusion [14]. For raw score

$s_j(q, d)$ from retriever j, the form used in this thesis is

$$\widetilde{s}_j(q, d) = \frac{s_j(q, d) - \min_{x \in C} s_j(q, x)}{\max\left(\max_{x \in C} s_j(q, x) - \min_{x \in C} s_j(q, x), \epsilon\right)},$$

where C is the indexed corpus and $\epsilon > 0$ prevents division by zero. The minimum and maximum are computed separately for each query and retriever. The resulting scores lie in $[0, 1]$. If all articles receive the same raw score, their normalized scores are zero. Normalization preserves each retriever's score ordering; it changes the scale used for fusion. These scores are used to rank articles and are distinct from evaluation metrics such as normalized discounted cumulative gain. Section 4.6 specifies the implemented normalization procedure.

An alternative is z-score normalization:

$$Z_j(q, d) = \frac{s_j(q, d) - \mu_j(q)}{\sigma_j(q)},$$

where $\mu_j(q)$ and $\sigma_j(q) > 0$ are the mean and standard deviation of the retriever's scores over the corpus. Z-scores can be negative and are not restricted to the range $[0, 1]$. This method is included as background; the reported experiments use Min-Max normalization. Both methods depend on the articles included in normalization, so the normalization set must be stated.

3.4.3 Weighted Score Fusion

Weighted score fusion adds the normalized retrieval scores after multiplying each by a weight [14]. The weights control how much each retriever contributes to the combined score. For a query q and an article d, $\widetilde{s}_b$, $\widetilde{s}_g$, and $\widetilde{s}_e$ denote the normalized scores from BM25, BGE-M3, and E5, respectively.

For BM25+BGE-M3, the combined score is

$$S_{BG}(q, d) = w_b \widetilde{s}_b(q, d) + (1 - w_b)\widetilde{s}_g(q, d), \qquad 0 \leq w_b \leq 1,$$

Here, w_b is the BM25 weight, and $1 - w_b$ is the BGE-M3 weight.

For BM25+BGE-M3+E5, the score becomes

$$S_{BGE}(q, d) = w_b \widetilde{s}_b(q, d) + w_g \widetilde{s}_g(q, d) + w_e \widetilde{s}_e(q, d),$$

The weights w_b, w_g, and w_e correspond to BM25, BGE-M3, and E5. They are nonnegative and satisfy $w_b + w_g + w_e = 1$. Articles are ranked by the combined score in descending order. Section 4.6 specifies the fusion procedure used in the experiments.

Example of Score Fusion. Consider the legal question:

> *"Is an employee entitled to severance allowance after terminating an employment contract?"*

BM25 and BGE-M3 independently assign retrieval scores to candidate legal articles. Their scores are normalized separately over the indexed corpus using the Min-Max procedure described above. The normalized scores are then combined using weights of 0.4 for BM25 and 0.6 for BGE-M3:

$$S_{BG}(q, d) = 0.4\widetilde{s}_b(q, d) + 0.6\widetilde{s}_g(q, d).$$

Table 3.1 shows hypothetical normalized scores for three candidate articles from the larger corpus. The article descriptions and weights are used only to illustrate the calculation.

Table 3.1: Illustrative BM25+BGE-M3 score fusion for three legal articles.

Legal article	**BM25**	**BGE-M3**	**Fused score**	**Final rank**
A: Severance allowance	0.90	0.95	**0.93**	**1**
B: Unemployment benefits	0.85	0.60	0.70	2
C: Contract termination	0.40	0.80	0.64	3

For Article A, the fused score is

$$S_{BG}(q, A) = 0.4 \times 0.90 + 0.6 \times 0.95 = 0.93.$$

The articles are ranked by fused score in descending order. Article A ranks first because both models assign it high scores. The example illustrates how fusion combines lexical matching from BM25 and embedding similarity from BGE-M3 into a single ranking score. It demonstrates the calculation, rather than establishing which provision legally answers the question.

3.5 Fine-Tuning and Negative Sampling

Pretraining on legal corpora develops language representations, as investigated in LEGAL-BERT [15]. Retrieval fine-tuning instead uses labeled questions and relevant passages to train their similarity scores [11]. This section explains the training objectives and negative examples used to fine-tune the BGE-M3 embedding model.

3.5.1 Contrastive Learning

BGE-M3 is fine-tuned to give relevant articles higher similarity scores than articles used as negative training examples. For a question q, d^+ is one article labeled as relevant, and

$N(q)$ is the set of negative training examples. The similarity between a question and an article is denoted by $s(q, d)$. A single-positive contrastive loss is [11]

$$\mathcal{L}(q) = -\log \frac{\exp(s(q, d^+)/\tau)}{\exp(s(q, d^+)/\tau) + \sum_{d^- \in N(q)} \exp(s(q, d^-)/\tau)}.$$

The loss decreases when the relevant article receives a higher score relative to the negative examples. The temperature $\tau > 0$ scales the similarity scores within the loss. Smaller values increase the effect of score differences.

A question may have several relevant articles among its training candidates. The multi-positive objective used in this thesis sums their exponentiated scores in the numerator and sums the exponentiated scores of all candidates in the denominator. This allows several relevant articles to contribute to the objective for one question.

The experiment that selects the model based on hybrid retrieval performance uses a single-positive loss. The experiment that selects the model based on standalone dense retrieval performance uses a multi-positive loss. Section 4.7 specifies the candidates and loss used in each experiment. A lower training loss must be followed by evaluation on a separate test set to assess retrieval performance.

3.5.2 Negative Sampling

Negative sampling selects articles that the model is trained to rank below the known relevant articles. Three common methods differ in how the examples are selected [11, 16]:

- **Random negatives:** Articles selected at random from the corpus.
- **In-batch negatives:** Articles from other training examples processed in the same batch. An article relevant to another question may be treated as a negative for the current question unless it is also identified as relevant to that question.
- **Hard negatives:** Articles ranked highly by a retriever but selected as negatives because they have no recorded positive label for the question.

An article without a positive label may still be relevant. Excluding known relevant articles from negative candidates reduces contradictory training labels but does not resolve missing annotations. The training procedure must also state how it treats other known relevant articles within the same batch. Section 4.7 describes the negative-selection rules and treatment of positive examples in the fine-tuning experiments.

This thesis uses hard negatives selected from the retriever's rankings. The next section explains how these examples are selected and filtered.

3.5.3 Hard-Negative Mining

Hard-negative mining selects highly ranked articles as negative training examples after excluding known relevant articles [16]. Randomly sampled articles may concern unrelated topics and be easy for the model to distinguish from the relevant article. Hard negatives provide more difficult comparisons: they may share vocabulary or meaning with the question and receive high retrieval scores without having a recorded positive label.

Training with these examples encourages BGE-M3 to distinguish the relevant article from similar articles that it may rank too highly. The objective is to assign the relevant article a higher similarity score, rather than treat all articles on the same topic as equally relevant. This helps the training focus on distinctions that random negatives may not provide. As noted above, an article without a positive label may still be relevant, so negative selection depends on the available annotations.

In the experiment that selects BGE-M3 based on hybrid retrieval performance, negatives are selected from BM25 and pretrained BGE-M3 rankings. Known relevant articles, duplicate candidates, unavailable articles, and articles from legal documents reserved for evaluation are excluded. The filter also excludes articles from the same legal document as the selected positive article. This reduces the risk of selecting a related article as a negative when its relevance is not recorded in the annotations. The negative groups are fixed before training. Section 4.7 describes their construction and filtering rules.

3.6 Target Matching and Evaluation Metrics

Evaluation checks whether ranked articles match the relevance targets recorded for each query. An article-level target requires both the document identifier and the article identifier to match. A whole-document target has an empty article identifier or the value `toanvan`; any article from that document can match it.

Results are checked in rank order against targets that have not yet been matched. Each target receives credit at most once. Thus, retrieving several articles from one document does not give repeated credit for a single whole-document target. If a retrieved article matches both an article-level target and a whole-document target, the evaluator's checking order determines which target receives credit.

In the formulas below, Q contains the evaluation queries, and R_q contains the distinct recorded targets for query q. The indicator $y_{q,r}$ is one when the article at rank r receives credit for a previously unmatched target, and zero otherwise. The notation @k means that only the first k ranked articles are considered.

Normalized discounted cumulative gain. Normalized discounted cumulative gain (NDCG) measures how well relevant results are placed near the top of a ranking [18]. A match at a higher rank contributes more than a match at a lower rank. The thesis uses NDCG@10 as the primary metric:

$$\mathrm{NDCG@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} / \log_2(r+1)}{\sum_{r=1}^{\min(|R_q|,10)} 1/\log_2(r+1)}.$$

The numerator sums the credit for matches, with a smaller contribution at lower ranks. The denominator gives the score of an ideal ranking that matches up to ten recorded targets at the highest positions. Queries without recorded targets receive zero. Targets unavailable in the corpus remain in the denominator, so the metric also reflects target availability and identifier mapping.

Hit and recall. Hit@k is one if at least one target is matched within the first k results, and zero otherwise:

$$\mathrm{Hit@}k(q) = \mathbb{1}\left[\sum_{r=1}^{k} y_{q,r} > 0\right],$$

Here, $\mathbb{1}[\cdot]$ equals one when its condition is satisfied and zero otherwise. The evaluator computes Hit@1, Hit@5, and Hit@10; the main results report Hit@10. Recall measures the proportion of recorded targets that are retrieved [1]. For a query with at least one target, it is

$$\mathrm{Recall@}k(q) = \frac{\sum_{r=1}^{k} y_{q,r}}{|R_q|}.$$

For example, matching one of two targets within the first k results gives Hit@$k = 1$ and Recall@$k = 0.5$. Hit records whether any target is found, while recall records how many of the targets are found. They coincide for queries with one target. Recall@100 is also reported for the standalone dense and hybrid comparisons in the fine-tuning experiment that selects the model using standalone retrieval performance (Tables 5.8 and 5.9).

Mean reciprocal rank. Mean reciprocal rank (MRR) measures how early the first target is matched. For a query whose first matching article occurs at rank r_q^*, the reciprocal rank (RR) at cutoff ten is

$$\mathrm{RR@10}(q) = \begin{cases} 1/r_q^*, & \text{if a target is matched within the first ten results,} \\ 0, & \text{otherwise.} \end{cases}$$

MRR@10 is the mean of these values across queries:

$$\mathrm{MRR@10} = \frac{1}{|Q|} \sum_{q \in Q} \mathrm{RR@10}(q).$$

For example, a first match at rank two gives a reciprocal rank of 0.5. Additional matches do not change this value because only the first matching rank is used.

Mean average precision. Average precision (AP) uses precision at each rank where a relevant result is retrieved. Mean average precision (MAP) averages AP across queries [1]. Here, precision $P_q(r)$ is the number of credited matches within the first r results divided by r. The evaluator calculates AP@10 and MAP@10 as

$$P_q(r) = \frac{1}{r} \sum_{j=1}^{r} y_{q,j},$$
$$\text{AP@10}(q) = \frac{\sum_{r=1}^{10} y_{q,r} P_q(r)}{\max(1, \min(|R_q|, 10))},$$
$$\text{MAP@10} = \frac{1}{|Q|} \sum_{q \in Q} \text{AP@10}(q).$$

AP@10 sums precision only at ranks that receive matching credit. Its denominator is the number of recorded targets, capped at ten, with a minimum of one to avoid division by zero. Queries without targets therefore receive zero. This is the evaluator's truncated definition; an implementation using the total number of targets as the denominator can give a different value when there are more than ten targets.

The reported query averages give each query equal weight, including queries with zero scores. When results are restricted to queries with available targets, that evaluation population is stated separately. All metrics use the recorded relevance targets, which may not include every legally relevant article.

3.7 Chapter Summary

This chapter explains how BM25 and dense retrieval models score legal articles, how score fusion combines their rankings, and how fine-tuning updates the BGE-M3 embedding model. It also defines the target-matching rules and evaluation metrics used to compare retrieval performance. Chapter 4 describes the experimental setup and evaluation procedures.

Chapter 4

Research Methodology

This chapter describes the research design, datasets, and implementation used to evaluate Vietnamese legal article retrieval. It presents corpus preparation and BCA dataset construction before the retrieval pipeline, score fusion, and BGE-M3 fine-tuning procedures. It then specifies the evaluation design, statistical analysis, experimental environment, and data-use limitations.

4.1 Research Design and Retrieval Task

This thesis uses controlled retrieval experiments to compare lexical retrieval, dense retrieval, and their combination for Vietnamese legal questions. The system receives a question, scores the indexed legal articles, and returns a ranked list. Each article has a legal document identifier and an article identifier. The primary outcome is normalized discounted cumulative gain at rank ten (NDCG@10); the matching rules and evaluation metrics are defined in Section 3.6.

BM25 provides the baseline for retrieval effectiveness. Comparing BM25+BGE-M3 with BM25 assesses the benefit of combining lexical and dense retrieval. Comparing BM25+BGE-M3+E5 with BM25+BGE-M3 assesses the additional benefit of multilingual-E5-base. Dataset and corpus conditions provide separate settings for these comparisons, with the same query population retained within each comparison of corpus versions. The evaluation also distinguishes articles absent from the corpus from errors in mapping target identifiers, since both can affect measured performance.

The fine-tuning experiments compare pretrained and fine-tuned BGE-M3 in standalone dense retrieval and in the hybrid configurations. One experiment selects the model using hybrid retrieval performance on development data; the other uses standalone dense retrieval performance. The hybrid comparisons keep BM25, E5, and the combination weights fixed while replacing BGE-M3 with its fine-tuned version. This design tests whether improvements in standalone retrieval are retained after score fusion. Section 4.7 describes the training procedures, and Section 4.8 specifies the evaluation conditions and their limitations.

Relevance labels support training, configuration selection, and evaluation. At retrieval time, the system uses only the question text and the indexed corpus. Published answers

and recorded relevance targets are not retrieval inputs. The experiments therefore measure recovery of annotated legal articles rather than the correctness of a generated legal answer.

4.2 Datasets and Corpus Preparation

The overall retrieval benchmark uses the supplied reference article text for ALQAC and Zalo. All methods within a dataset use the same ordered queries, corpus, and relevance targets. BCA uses the separately constructed corpus described in Section 4.3. The completed Zalo fine-tuning experiments use their recorded training and evaluation corpora, described below and in Section 4.7; their results are separate from the reference-text benchmark.

Table 4.1 summarizes the three baseline datasets. Queries that share legal references are grouped for selection and evaluation, as explained in Section 4.8.

Table 4.1: Baseline datasets.

Dataset	**Queries**	**Articles**	**Evaluation design**
ALQAC	811	3,363	Grouped cross-validation using reference article text
Zalo	3,196	61,425	Grouped cross-validation using reference article text
BCA	546	14,789	Separate evaluation of the largest linked group and grouped cross-validation over smaller groups

4.2.1 ALQAC and Zalo Datasets

ALQAC and Zalo provide Vietnamese legal questions with annotations identifying the relevant legal articles. The questions serve as retrieval queries, and the annotated references provide the targets used to evaluate the ranked results.

For ALQAC, the baseline evaluation combines the supplied training and private-test questions, giving 811 queries. Its reference corpus contains 3,363 articles.

For Zalo, the baseline evaluation uses 3,196 queries and a reference corpus of 61,425 articles. Each query retains its supplied relevance targets.

For both datasets, questions linked by shared legal references are kept in the same group. Fusion weights are selected using other groups before evaluation on the excluded group, as described in Section 4.8. All baseline methods within each dataset use the supplied reference corpus and annotations. The following subsection distinguishes this corpus from the preparation used in the completed fine-tuning experiments.

4.2.2 Article Extraction and Identifier Alignment

Each article is stored with its text, document identifier, and article identifier. The ALQAC and Zalo benchmark uses the text supplied with each dataset.

For the Zalo fine-tuning comparisons, article text is extracted from legal documents collected in raw HTML format, then cleaned and processed. The text is split into articles and matched to the supplied records by document and article identifiers. Extracted text replaces the supplied text when a match is found and the cleaned content is nonempty. Otherwise, the supplied text is retained. This reference fallback preserves all 61,425 articles for the pretrained and fine-tuned comparisons.

Identifiers are trimmed and converted to lowercase before target matching. All retrievers use the same article order so that their scores remain linked to the correct articles.

4.3 BCA Dataset Construction

The BCA dataset is constructed from public questions and published answers in the question-and-answer section of the Bộ Công an portal.[1] Each question becomes a retrieval query, and legal references extracted from its published answer become the recorded relevance targets. Construction consists of collecting the source records, cleaning their text, extracting legal references, and mapping those references to corpus identifiers. The answer text is used to construct targets and is not supplied to the retriever.

4.3.1 Motivation and Research Value

The first purpose is to extend evaluation to questions submitted through a public-service channel. Competition benchmarks provide defined tasks and evaluation conditions; for example, ALQAC evaluates legal retrieval and question answering using a manually annotated dataset [8]. BCA adds questions concerning residence, identity documents, fire prevention, traffic penalties, and immigration. Citizens may describe their circumstances in everyday language rather than use the terms found in legislation. This motivates examining retrieval on public-service questions alongside the ALQAC and Zalo benchmark comparisons.

The second purpose is to construct a research resource beyond the supplied benchmarks. The collection and reference-extraction pipeline converts published question-answer records into queries and traceable relevance targets. It supports evaluation of BM25, BGE-M3, E5, and their combinations on a separately collected set of questions. The resulting analysis examines performance across collections, including differences in target availability and

[1] https://bocongan.gov.vn/hoi-dap/

shared legal references. The resource contribution is the constructed collection and its documented processing procedure.

4.3.2 Data Sources and Record Selection

The collection script requests successive pages of question listings through the portal's application programming interface, then retrieves each question and its published answer. Each record is assigned an identifier derived from the source question identifier. During a repeated collection, records with existing identifiers are skipped to prevent duplication.

Acquisition requires separate listing and detail requests: the listing identifies available questions, while each detail response supplies the question and answer fields. Failed listing requests can interrupt discovery, and failed detail requests can leave individual records uncollected.

The saved collection contains 652 records with distinct question identifiers. Of these, 546 are retained for retrieval evaluation because they contain at least one extracted legal document reference.

4.3.3 Text Cleaning and Reference Extraction

Preprocessing covers the collected question-answer records, extracted legal articles, and inputs to each retriever. Legal references are then derived from the published answers to construct evaluation targets. The question text remains the retrieval query throughout this process.

Text cleaning and retriever inputs. The preprocessing steps are organized by their role:

1. **Cleaning question-answer records.** The response fields `content` and `answer` contain the question and published answer. Regular expressions replace HTML tags with spaces, remove leading and trailing whitespace, and reduce repeated spaces, tabs, and line breaks to a single space. Unicode Normalization Form C (NFC) standardizes equivalent character sequences in Vietnamese text. The cleaned question is used as the retrieval query, while the cleaned answer is used for reference extraction.

2. **Extracting legal articles.** Legal HTML is converted to text lines using BeautifulSoup. Article boundaries are identified from lines beginning with "Điều" and its alternative spelling, followed by an article number and an optional letter suffix. The leading article marker and number are removed where present.

3. **Preparing BM25 tokens.** Question and article text is converted to lowercase. Punctuation and other characters outside the word-character and whitespace classes

are replaced with spaces, and the text is split on whitespace. Vietnamese letters with diacritics are retained. This procedure uses no external Vietnamese word-segmentation library.

4. **Formatting dense inputs.** BGE-M3 receives question and article text without additional prefixes. Multilingual-E5-base receives the prefix `query:` for questions and `passage:` for articles. Both models use a maximum input length of 512 tokens in the baseline experiments. Section 4.5 describes the retrieval models, and Section 4.7 specifies the separate fine-tuning configurations.

Legal reference extraction. The thesis processes the published answer using the following steps:

1. **Identifying articles and documents.** The answer is divided into spans using punctuation and line boundaries. Within each span, the extractor identifies article numbers, including forms such as "Điều 5", "Điều 12a", and "Điều 32". It also detects document terms such as "Luật", "Bộ luật", "Nghị định", and "Thông tư".

2. **Filtering ambiguous phrases.** Keywords such as "quy định", "quyết định", and "nghị quyết" require a following number to be treated as document references. Ordinary expressions such as "luật sư" and "luật chơi" are excluded. The rules also filter references marked as drafts and remove trailing phrases, incomplete parentheses, and other text outside the detected document name. Document-specific exceptions handle cases not covered by the general rules.

3. **Linking articles to documents.** The extractor first attempts to link an article mention to the following document mention when the intervening words are permitted connectors. If that association fails, it checks the preceding document using a separate connector-word set. This handles both "Điều 9 Nghị định 79/2014/NĐ-CP" and "Nghị định số 79/2014/NĐ-CP tại Điều 9, Điều 15". A document cited without a specific article is recorded with an empty article identifier, representing a whole-document target.

4. **Removing duplicate references.** Repeated document and article pairs are removed. When two extracted names refer to the same article and one name is contained in a longer version, the more complete name is retained. For example, "Luật Giao thông đường bộ năm 2008" is retained over the shorter "Luật Giao thông đường bộ" when both have been extracted for the same article.

The extracted references are saved with the cleaned question-answer record. Section 4.3.4 describes their mapping to corpus identifiers and the checks for target availability. Eval-

uation uses the matching rules in Section 3.6: a whole-document target can match any indexed article from that document, and each recorded target receives credit at most once. Figure 4.1 summarizes the construction process and its connection to corpus coverage checks.

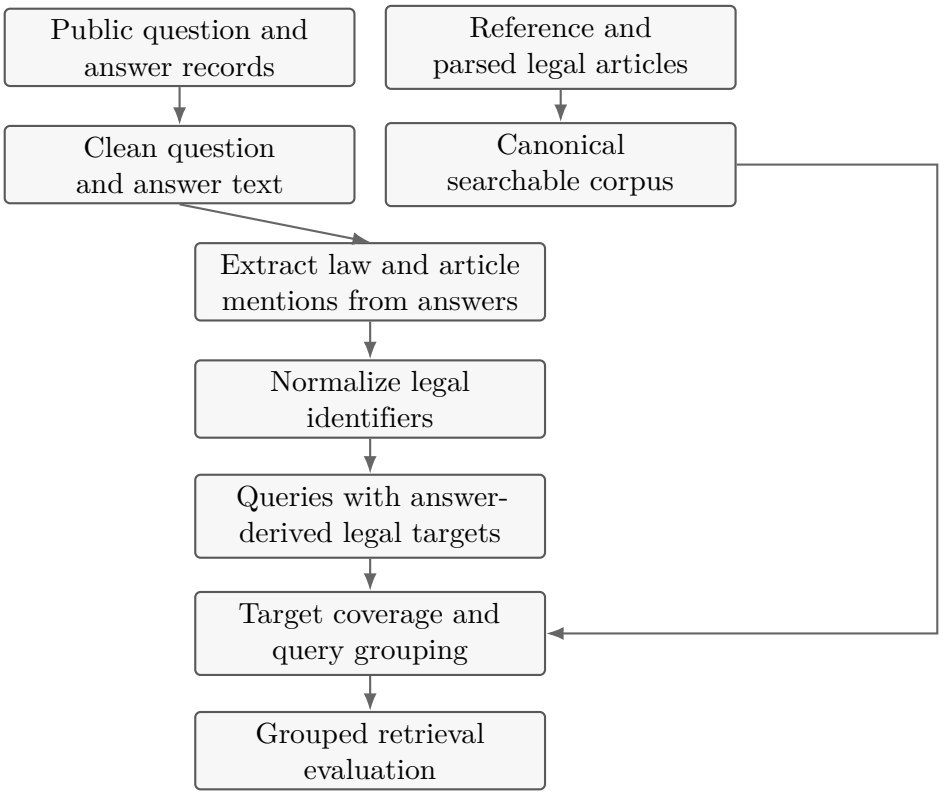


Figure 4.1: Construction of the BCA retrieval collection. Published answers supply reference evidence; only question text is used as the retrieval query.

4.3.4 Target Mapping and Corpus Coverage

Extracted legal references are mapped to corpus identifiers to construct evaluation targets. A correction table associates extracted document names with document identifiers. For example, "Nghị định số 100/2019/NĐ-CP" is mapped to `100/2019/nd-cp`. Leading and trailing whitespace is removed from document and article identifiers, and both are converted to lowercase. Each target consists of a document identifier and an article identifier. The article identifier is left empty when the published answer cites a document without specifying an article. These normalization steps standardize identifiers but do not verify that the extracted reference has been mapped to the correct legal document.

The reference corpus is extended with legal documents collected from public sources in raw HTML format. Article text is extracted from these documents and added to the corpus, while the raw HTML and source mappings are saved. Duplicate article identifiers within each document and articles with empty text are excluded. The resulting searchable corpus

contains 14,789 articles.

References that remain unresolved may be assigned the placeholder `INTERNAL_DOCUMENT`. Entries with this placeholder are excluded from corpus additions, but their questions and recorded targets remain in the evaluation set. The placeholder indicates an unresolved mapping, it does not establish that the referenced document is internal or access-restricted.

Target coverage is checked against the indexed article identifiers using the matching rules in Section 3.6. An article-level target requires a match on both the document and article identifiers. A whole-document target requires at least one indexed article with the same document identifier. A question has full coverage when all its recorded targets have a match, partial coverage when only some have a match, and no coverage when none has a match. These categories measure coverage of the recorded targets under the matching rules, rather than completeness of the corpus for answering the legal question.

A target without a match can result from missing corpus content or an unresolved or incorrect mapping. If the referenced article is absent from the corpus, no ranking method can retrieve it. If its text is present but the target is mapped incorrectly, retrieval may recover the relevant article without receiving evaluation credit. For a correctly mapped target present in the corpus, failure to retrieve a match within the evaluation cutoff concerns ranking performance. Therefore, coverage checks alone cannot establish whether a retrieval failure is caused by missing content, incorrect mapping, or ranking.

4.3.5 Example of Query and Target Construction

Table 4.2 illustrates how a question about fire-prevention equipment in apartment buildings becomes a retrieval query with recorded relevance targets. The published answer cites Nghị định số 79/2014/NĐ-CP. Reference extraction produces one whole-document target and two article-level targets, for Articles 9 and 15. The document identifier is normalized according to Section 4.3.4, while the question text supplies the retrieval query.

Table 4.2: Worked BCA question-to-target transformation.

Stage	**Representation**	**Interpretation**
Stored query	"Bộ Công an cho tôi hỏi có quy định bắt buộc nào về các trang thiết bị, dụng cụ phòng cháy chữa cháy đối với nhà chung cư không?"	Question text supplies the retrieval input

Continued on next page

Stage	Representation	Interpretation
Extracted answer references	Nghị định số 79/2014/NĐ-CP, with article identifiers empty, 9, and 15	The saved extraction contains one whole-document reference and two article references
Normalized document identifier	`79/2014/nd-cp`	The mapped identifier is converted to lowercase
Evaluation targets	`(79/2014/nd-cp, empty)` `(79/2014/nd-cp, 9)` `(79/2014/nd-cp, 15)`	Each pair contains a document identifier and an article identifier; `empty` denotes an empty article identifier
Retrieval inputs	Question text as the query; indexed article text as the corpus	The published answer is used only to construct targets, which are used for evaluation

The empty article identifier allows any indexed article from this decree to match the whole-document target. The other two targets require the corresponding article identifiers. Under the rules in Section 3.6, each retrieved article receives credit for at most one previously unmatched target. When it could match both a whole-document target and an article-level target, the evaluator's target-checking order determines which target receives credit. Retrieving several articles from the decree does not repeatedly earn credit for the whole-document target.

4.4 Overview of Retrieval Experiments

The retrieval experiments use Vietnamese questions as queries and legal article text as the searchable corpus. BM25 computes lexical scores, while BGE-M3 and multilingual-E5-base, referred to as E5, compute dense similarity scores. Each retriever scores every article in the same corpus order, ensuring that scores combined across retrievers refer to the same article. The experiments evaluate BM25, BGE-M3, and E5 individually, as well as the hybrid configurations BM25+BGE-M3 and BM25+BGE-M3+E5.

For hybrid retrieval, each retriever's scores are normalized separately for each query over the full corpus. The normalized scores are combined using a weighted sum, and articles are ranked by the combined score. Fusion weights are selected on the weight-selection queries and fixed before ranking the separate evaluation queries. The system returns the ten highest-ranked articles with their document and article identifiers. The returned identifiers are compared with the recorded relevance targets to compute NDCG@10 and the other evaluation metrics. Section 4.6 specifies score normalization and fusion, and Section 4.8

describes query grouping and the allocation of queries to weight selection and evaluation. Figure 4.2 summarizes this procedure with pretrained dense models.

The fine-tuning comparisons replace pretrained BGE-M3 with a fine-tuned model while keeping BM25, E5, and the corresponding fusion weights fixed. Standalone dense retrieval is evaluated separately to examine whether changes in BGE-M3's retrieval performance are also observed in the hybrid configurations. Section 4.7 describes the training and model selection procedures. The evaluated system produces article rankings; reranking and answer generation are outside its scope.

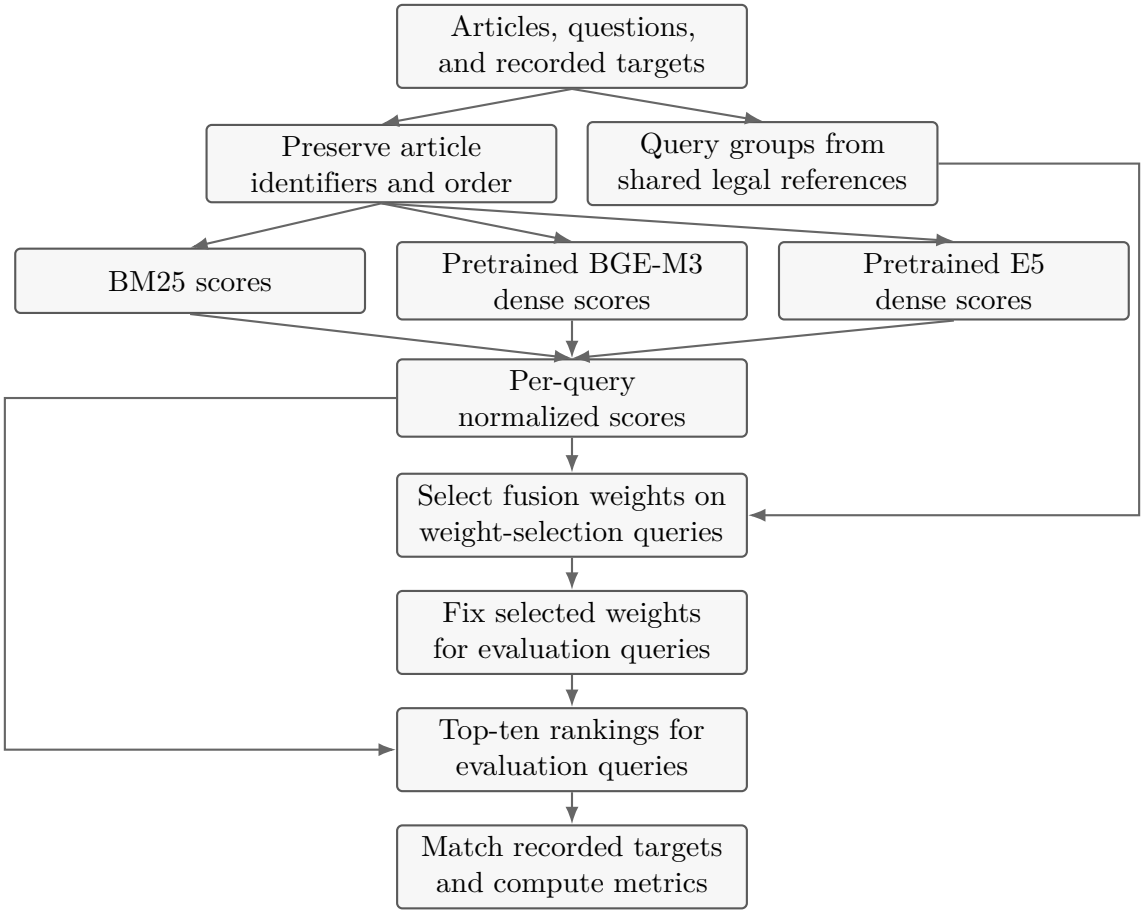


Figure 4.2: Hybrid scoring and evaluation pipeline with pretrained dense models. Scores are computed over the full corpus. Recorded targets support query grouping; targets for the weight-selection queries guide fusion-weight selection, and targets for the separate evaluation queries are used to compute metrics. Retrievers receive question and article text.

4.5 Retrieval Models and Implementation

The experiments use BM25 for lexical retrieval and the dense representations of BGE-M3 and multilingual-E5-base for semantic retrieval. All models score the same ordered articles

within each corpus version. Preserving article order allows their scores to be combined without changing the associated legal document and article identifiers.

4.5.1 BM25

BM25 scores tokenized question and article text. Preprocessing converts text to lowercase, replaces non-word punctuation with token separators, and splits the resulting text on whitespace. This preprocessing remains fixed in the reported comparisons. The experiments therefore evaluate this BM25 configuration rather than alternative Vietnamese tokenization methods.

4.5.2 BGE-M3 and Multilingual-E5-base

BGE-M3 is the primary dense retriever. Multilingual-E5-base, referred to as E5, provides an additional dense score for the BM25+BGE-M3+E5 configuration. The experiments use BGE-M3's dense representation; its sparse and multi-vector retrieval methods are not evaluated.

Each encoder maps the question and each article to vectors. The implementation divides each vector by its Euclidean norm, with a minimum denominator of 10^{-10}, and computes the dot product of the normalized vectors. For the query vector $\mathbf{u}_q$ and article vector $\mathbf{v}_d$ before normalization, the resulting score is

$$s_{\text{dense}}(q, d) = \frac{\mathbf{u}_q^\top \mathbf{v}_d}{\max(\|\mathbf{u}_q\|_2, 10^{-10}) \max(\|\mathbf{v}_d\|_2, 10^{-10})}.$$

Here, q is the question, d is an indexed article, and $\|\cdot\|_2$ denotes the Euclidean norm. For vectors whose norms exceed the numerical threshold, this calculation is cosine similarity. The threshold prevents division by zero.

Table 4.3 summarizes the embedding model settings used in the baseline benchmark. Both models use their pretrained parameters without further training. The separate BGE-M3 fine-tuning configurations are described in Section 4.7.

The evaluation scores every indexed article rather than restricting retrieval to candidates from an approximate nearest-neighbor index. Thus, an available target that falls outside the first ten results is excluded by the resulting ranking rather than by a separate candidate filter. This exhaustive scoring design supports comparisons of ranking performance; it does not establish the system's latency in deployment.

Table 4.3: Embedding model settings for baseline retrieval.

Setting	**BGE-M3**	**multilingual-E5-base**
Model identifier	`BAAI/bge-m3`	`intfloat/multilingual-e5-base`
Embedding dimension	1,024	768
Query prefix	None	`query:`
Article prefix	None	`passage:`
Maximum input length	512 tokens	512 tokens
Vector normalization	L2 normalization	L2 normalization
Similarity score	Dot product of normalized vectors	Dot product of normalized vectors

4.6 Score Normalization and Fusion

BM25 and dense retrieval produce scores on different scales. Before combining them, the baseline applies Min-Max normalization separately to each retriever's scores for each query. For retriever r, raw score $s_r(q, d)$, and full indexed corpus C, the normalized score is

$$\widetilde{s}_r(q, d) = \frac{s_r(q, d) - \min_{d' \in C} s_r(q, d')}{\max\left(\max_{d' \in C} s_r(q, d') - \min_{d' \in C} s_r(q, d'), \epsilon\right)}.$$

Here, ϵ is a small positive constant that prevents division by zero. If a retriever assigns the same score to every article for a query, all its normalized scores are zero. Normalization uses the full corpus rather than a shortlist of retrieved candidates, so every indexed article can enter the final ranking.

The BM25+BGE-M3 score is

$$S_{bg}(q, d) = w_b \widetilde{s}_b(q, d) + (1 - w_b)\widetilde{s}_g(q, d),$$

where b denotes BM25, g denotes BGE-M3, and $0 \leq w_b \leq 1$. Adding E5 gives

$$S_{bge}(q, d) = w_b \widetilde{s}_b(q, d) + w_g \widetilde{s}_g(q, d) + w_e \widetilde{s}_e(q, d),$$
$$w_b + w_g + w_e = 1,$$

where e denotes E5 and all weights are nonnegative. The articles are ranked by the combined score in decreasing order.

The baseline selects weights from a grid with increments of 0.1. BM25+BGE-M3 has 11 candidate settings, with $w_b \in \{0, 0.1, \ldots, 1\}$. BM25+BGE-M3+E5 has 66 candidate settings whose weights sum to one. This grid includes zero-weight settings, including $w_e = 0$, so it can reproduce every BM25+BGE-M3 setting in the baseline grid. A higher score on the weight-selection queries therefore does not establish that E5 improves retrieval

on separate evaluation queries.

For each baseline split, the weights with the highest mean NDCG@10 on the weight-selection queries are fixed before evaluation on the other queries. Section 4.8 describes the grouping and fold procedure. Algorithm 4.1 shows how the selected configuration produces a ranking for one query.

Algorithm 4.1 Article ranking with fixed fusion weights

Require: Query q, corpus C, retriever scores, fixed fusion weights w
Ensure: Ten highest-ranked articles
1: Normalize each retriever's scores over the full corpus for q
2: **for** each article $d \in C$ **do**
3: combined$[d] \leftarrow \sum_r w_r$ normalized_score$[r, d]$
4: **end for**
5: **return** the ten articles with highest combined score

The fine-tuning comparisons use weights selected with pretrained BGE-M3 and keep them fixed when replacing it with the fine-tuned model. BM25 and E5 also remain unchanged. These comparisons measure the effect of replacing BGE-M3 within the specified hybrid configuration. The 0.1 grid described above applies to baseline selection. The fine-tuning experiments use separate selection procedures, described in Section 4.7.

4.7 BGE-M3 Fine-Tuning

The two fine-tuning experiments differ in how they select BGE-M3. The first selects the model using hybrid retrieval performance on development queries, followed by evaluation on a separate test set. The second selects it using standalone dense retrieval performance, then evaluates the selected model both on its own and in BM25+BGE-M3 and BM25+BGE-M3+E5. Within each hybrid comparison, fine-tuned BGE-M3 replaces pretrained BGE-M3 while BM25, E5, and the selected fusion weights remain fixed. These comparisons test whether fine-tuning improves standalone retrieval and whether any improvement is retained in the hybrids under fixed fusion weights. The experiments use different training procedures and are therefore described separately. Section 5.3.2 reports the results and evaluation limitations.

4.7.1 Model Selection Based on Hybrid Retrieval

This experiment fine-tunes BGE-M3 on the Zalo dataset and selects its training settings using hybrid retrieval performance on development queries.

Training Data and Negative Examples

For each eligible question, one article labeled as relevant is selected. The article must be available in both the reference-text and extracted-text corpus versions. The question is paired with a article to create two training groups. Each group contains one positive article and seven explicit negative articles.

Negative candidates are taken from up to 256 of the highest-ranked results of each of BM25 and pretrained BGE-M3 using the extracted-text corpus. Filtering removes known relevant articles, duplicate candidates, unavailable articles, and articles from legal documents reserved for development or test evaluation. Articles from the same legal document as the positive are also excluded to reduce the risk of selecting a related but unannotated article as a negative.

Training Objective and Configuration

The training objective is to make BGE-M3 assign a higher similarity score to the selected relevant article than to the other articles in the batch. Each question is supplied with one positive article and seven explicit negatives. Articles from other question groups in the same batch provide additional comparisons. Only BGE-M3's parameters are updated during this training.

A batch contains B question groups, with $G = 8$ articles per group. The normalized vectors for question i and article j are $\mathbf{q}_i$ and $\mathbf{p}_j$. The selected positive article is at position $y_i = iG$, because it is stored first in each group. The training loss is

$$\mathcal{L}_{\text{batch}} = -\frac{1}{B}\sum_{i=0}^{B-1}\log\frac{\exp(\mathbf{q}_i^\top \mathbf{p}_{y_i}/\tau)}{\sum_{j=0}^{BG-1}\exp(\mathbf{q}_i^\top \mathbf{p}_j/\tau)}.$$

The dot product measures similarity, and $\tau = 0.02$ scales these scores before the loss is calculated. The numerator uses the selected positive article, while the denominator includes all articles in the batch. The loss decreases as the positive article's score increases relative to the others. The final loss is averaged over the B questions.

With the selected batch size of four question groups, each question is compared with 32 article vectors. Gradients are accumulated over eight forward passes before updating the model, so one update uses 32 question groups. Each loss calculation still compares only the articles within its own forward pass.

Table 4.4 lists the tested hyperparameter values and the selected configuration. The full training configuration is recorded in Appendix D.

Table 4.4: Hyperparameter search for hybrid-based model selection.

Parameter	Values tested	Selected configuration
Contrastive query batch size	$\{1, 2, 4\}$	**4**
Learning rate	$\{5 \times 10^{-6}, 1 \times 10^{-5}, 2 \times 10^{-5}\}$	$\mathbf{2 \times 10^{-5}}$
Training epochs	$\{1, 2, 3\}$	**2**

Model Selection and Evaluation

Table 4.5 shows the allocation of the 3,196 Zalo queries during model selection.

Table 4.5: Zalo fold allocation for hybrid-based model selection.

Fold	Queries	Role
0	**640**	Test
1	**639**	Development for configuration selection
2	**639**	Pilot training
3	**639**	Pilot training
4	**639**	Pilot training
Total	**3,196**	

Models are trained on folds 2, 3, and 4, then evaluated on fold 1 to select the learning rate, epoch count, and query batch size. Selection uses BM25+BGE-M3 NDCG@10. Compared with the pretrained baseline, the development rule requires an increase of at least 0.01 for BM25+BGE-M3 and allows a decrease of at most 0.01 for BM25+BGE-M3+E5.

The selected configuration is then used to train final models on folds 1, 2, 3, and 4 with seeds 42, 43, and 44. Fold 0 is the test dataset. Combination weights selected on folds 1, 2, 3, and 4 using pretrained BGE-M3 are kept fixed during the test comparison, as are BM25 and E5.

Figure 4.3 summarizes the selection and evaluation sequence.

4.7.2 Model Selection Based on Standalone Dense Retrieval

Training Data and Configuration

This experiment selects BGE-M3 using standalone dense retrieval before testing it in BM25+BGE-M3 and BM25+BGE-M3+E5. It uses a fixed corpus of 61,425 parsed articles, with reference text retained where parsed text is unavailable. The official training partition contains 2,556 queries. Removing one near-duplicate of an official test query leaves 2,555 eligible training groups. The final evaluation uses 640 official test queries.

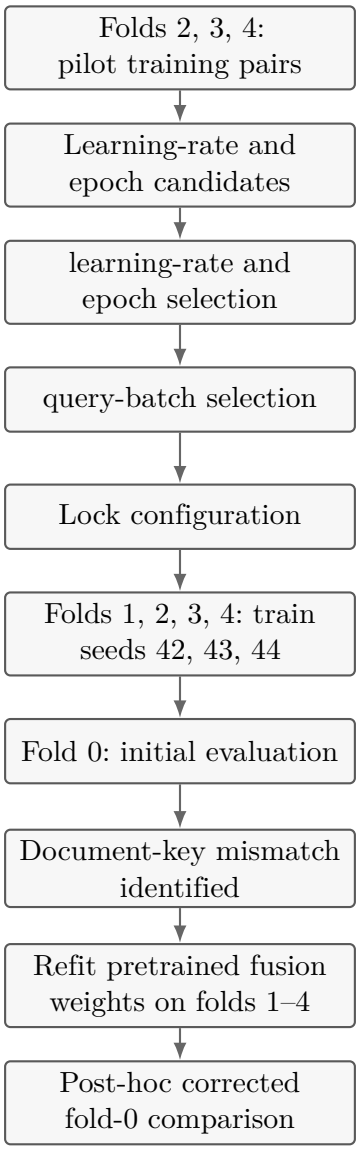


Figure 4.3: Model selection and test evaluation.

Each group includes five explicit negatives. The training procedure uses a mask to identify all known positive articles within the batch candidate set. This differs from the selected-positive objective in Section 4.7.1. The encoder uses the hidden state at the first token position, L2 normalization, and query and passage limits of 256 and 512 tokens, respectively. The specified temperature is 0.05, with bfloat16 precision, a cosine learning-rate schedule, and a warmup ratio of 0.05. The final runs use a contrastive batch of 64 queries with gradient caching and chunks of two sequences, as shown in Figure 4.4.

Training Objective

Utilising hard negatives, the training objective is to maximize the similarity score of each query with its known positive articles while minimizing the score with all other articles in the batch. The experiment loop thought configuration finetune to find the best model based on the highest NDCG@10 on the development set. The selected model is then evaluated on the official test queries, both as a standalone dense retriever and within the hybrid configurations.

Model Selection and Hybrid Evaluation

BGE-M3 is selected using standalone NDCG@10 on development data, with Recall@100 checked to limit losses in target recovery. This evaluates the embedding model's retrieval performance before it is combined with BM25 and E5.

The selected model is then evaluated in BM25+BGE-M3 and BM25+BGE-M3+E5 on the same test dataset used for the standalone comparison. Fusion weights are selected using pretrained BGE-M3 on training queries. These weights, BM25, and E5 remain fixed when fine-tuned BGE-M3 replaces the pretrained model. This comparison tests whether a standalone improvement is retained after score fusion. Section 5.3.2 reports the results and evaluation limitations.

4.8 Evaluation Design and Statistical Analysis

4.8.1 Query Grouping and Data Splits

Questions that cite the same legal document are kept in one group. A question citing several documents links their groups together. This prevents questions linked by shared legal references from being split between weight selection and evaluation.

For the pretrained ALQAC and Zalo comparisons, one fold is evaluated using fusion weights selected on the remaining folds. Repeating this procedure gives each question one evaluation result without using its labels to select the weights applied to it. The query

populations are defined in Section 4.2.

Algorithm 4.2 summarizes the procedure. Metrics are averaged over all evaluation queries, giving each query equal weight. The selected weights may differ between folds.

Algorithm 4.2 Grouped selection and evaluation of fusion weights

Require: Ordered queries Q, corpus D, aligned normalized score matrices, grouped folds, ordered candidate weight grid W
Ensure: Selected weights per fold and evaluation metrics for every query
for each fold f **do**
    Use the other folds for selection (T) and fold f for evaluation (V)
    for each candidate weight vector $w \in W$, in recorded order **do**
        Rank D for every query in T using the weighted score sum
        Compute mean query NDCG@10
    end for
    Select the first candidate with the highest selection mean
    Rank all queries in V using the selected weights
    Store each evaluation query's metrics
end for
Aggregate evaluation-query rows across folds, giving each query equal weight

The fine-tuning experiments use the training, development, and test allocations in Section 4.7. Each fine-tuned model is compared with its matched pretrained baseline on the same test dataset.

4.8.2 Evaluation Metrics and Comparisons

NDCG@10 is the primary metric because it rewards matches at higher ranks. Hit@10 records whether any target is retrieved, MRR@10 measures how early the first match occurs, and MAP@10 summarizes precision at matching ranks. The experiment selected using standalone dense performance also reports Recall@100 for its standalone and hybrid comparisons. Section 3.6 defines the matching rules and formulas.

The evaluation separates three comparisons:

- **Hybrid retrieval versus BM25:** compare BM25+BGE-M3 with BM25 to assess the benefit of combining lexical and dense retrieval.
- **Contribution of E5:** compare BM25+BGE-M3+E5 with BM25+BGE-M3 to assess the additional contribution of E5.
- **Effect of fine-tuning:** compare pretrained and fine-tuned BGE-M3, both on its own and within the hybrids. Within each hybrid comparison, BM25, E5, and the combination weights remain fixed.

A practically useful hybrid improvement requires an absolute NDCG@10 increase of at least 0.01. Statistical uncertainty is assessed separately.

4.9 Experimental Environment and Reproducibility

The experiments run on Google Colab with a single NVIDIA L4 GPU. A local machine launches the runtime and monitors its logs through the Google Colab command-line interface. Figure 4.4 shows the execution environment for the fine-tuning experiment that selects BGE-M3 using standalone dense retrieval.

Code and data are copied from Google Drive to the runtime disk and monitor task progress. Training runs as a background Python process, with seeds 42, 43, and 44 executed sequentially for 60 optimizer steps each. Gradient caching supports a contrastive batch of 64 queries while processing chunks of two sequences.

Checkpoints are saved at steps 30 and 60, verified, reloaded, and copied to Google Drive. The runtime disk holds temporary files, while Google Drive retains the files needed to resume from a valid checkpoint after a runtime interruption.

The experiments assess retrieval effectiveness. They do not provide controlled measurements of query latency, throughput, or deployment cost, so the recorded hardware specifications do not support conclusions about those outcomes.

4.10 Research Ethics and Data Use

The experiments use the ALQAC and Zalo collections and the BCA dataset described in Sections 4.2 and 4.3. BCA is derived from public questions and published answers on the Bộ Công an portal. Public access identifies the source of these records, but does not by itself establish permission for unrestricted reuse or redistribution. The construction records described in this thesis do not establish consent from the people who submitted the questions or document a completed review of personal information.

Public questions and answers may contain details that identify individuals. Any future release of BCA therefore requires a review of identifying information, source terms, and the permitted scope of redistribution. The example in Section 4.3 summarizes the query and target structure without reproducing personal details. Published answers are retained as evidence for reference extraction and are not supplied as query inputs during retrieval.

The evaluation measures recovery of recorded legal references. These references are extracted from published answers and may be incomplete or incorrectly mapped, as discussed in Section 4.3. Retrieval scores therefore do not establish the correctness of a legal answer or the current applicability of a retrieved provision. The experiments assess article ranking and do not evaluate a system for providing legal advice.

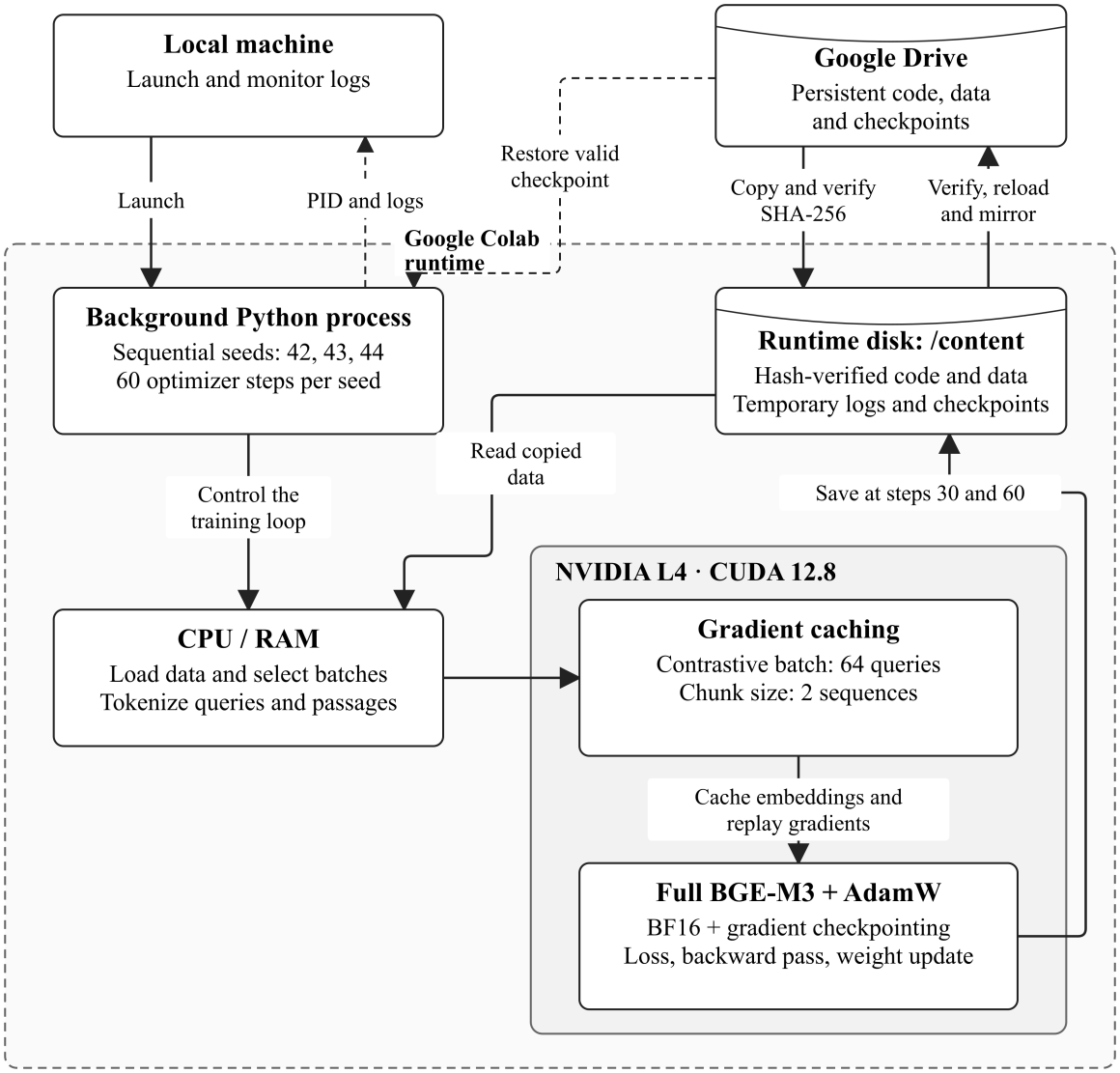


Figure 4.4: Execution environment for standalone BGE-M3 fine-tuning.

4.11 Chapter Summary

This chapter presents the research design, the datasets and corpus preparation, and the construction of BCA from public questions and published answers. It specifies lexical and dense retrieval, score normalization and fusion, and BGE-M3 fine-tuning with model selection based on either hybrid or standalone dense retrieval. The evaluation design explains query grouping, data splits, metric comparisons, and the scope of statistical analysis. The chapter also documents the experimental environment, reproducibility requirements, and data-use limitations; Chapter 5 reports the results.

Chapter 5

Experiments and Evaluation

This chapter describes the experimental conditions and benchmark evaluation procedure, then presents the results and analysis. The results cover the Bộ Công an (BCA) collection, BGE-M3 fine-tuning relative to matched pretrained baselines, and the overall retrieval benchmark, including the contributions of hybrid retrieval and multilingual-E5-base (E5).

5.1 Experimental Conditions

The benchmark uses the dataset ALQAC and Zalo, while BCA uses its separately constructed corpus. The retrieval models are BM25, `BAAI/bge-m3`, and `intfloat/multilingual-e5-base`. Baseline dense inputs are limited to 512 tokens. Dataset sizes, preprocessing, and model settings are specified in Sections 4.2 and 4.5.

The experiment were ran on a single NVIDIA L4 GPU on Google Colab platform. The reason is that the L4 GPU is a powerful and cost-effective option for training and fine-tuning machine learning models. It provides sufficient computational resources to handle the training of large language models like BGE-M3 and E5, while also being more accessible and affordable compared to higher-end GPUs. Additionally, Google Colab offers a convenient environment for running experiments without the need for extensive local hardware setup, making it an ideal choice for this research.

5.2 Benchmark & Evaluation Procedure

The overall benchmark uses the article text supplied with the ALQAC and Zalo datasets. BCA is evaluated using the corpus constructed from collected legal documents. The fine-tuning experiments are evaluated separately, each comparing pretrained and fine-tuned BGE-M3 on the same corpus and test dataset. Sections 4.2 and 4.3 describe the datasets and corpus preparation.

The evaluation follows six steps:

1. **Prepare the inputs.** Align question text, article text, and relevance targets by their document and article identifiers. All retrievers use the same article order within each comparison. Only the question text is supplied as the query.

2. **Define the evaluation splits.** For the pretrained benchmark, keep questions linked by shared legal references in the same group. For ALQAC and Zalo, evaluate each fold using fusion weights selected on the remaining folds. Fine-tuning uses the training, development, and test allocations in Section 4.7.

3. **Score and rank the articles.** BM25, BGE-M3, and E5 score every article in the corpus. Each model ranks articles by its own scores. For hybrid retrieval, apply Min-Max normalization separately to each model's scores for each query, then rank articles by their weighted sum (Section 4.6).

4. **Select weights and fine-tuned models.** Select fusion weights using mean normalized discounted cumulative gain at rank ten (NDCG@10) on the selection queries. Select fine-tuned BGE-M3 using development performance in either hybrid or standalone dense retrieval, according to the experiment. For the fine-tuning comparisons, select fusion weights with pretrained BGE-M3 and keep them fixed when evaluating the fine-tuned model.

5. **Calculate evaluation metrics.** Match ranked articles to the recorded targets using the rules in Section 3.6. NDCG@10 is the primary metric. Hit@10 records whether a target appears in the first ten results. Mean reciprocal rank at ten (MRR@10) measures how early the first match occurs, while mean average precision at ten (MAP@10) accounts for precision at each matching rank. The experiment selected using standalone dense retrieval also reports Recall@100 for its standalone and hybrid comparisons.

6. **Compare retrieval performance.** Compare BM25+BGE-M3 with BM25 to assess improvement over the baseline, and BM25+BGE-M3+E5 with BM25+BGE-M3 to assess the contribution of E5. Evaluate fine-tuning by comparing pretrained and fine-tuned BGE-M3 on the same test dataset. Assess standalone and hybrid results separately, keeping BM25, E5, and fusion weights fixed within each hybrid comparison.

The ALQAC and Zalo benchmark averages the evaluation scores of all queries, giving each query equal weight. Each query is evaluated once, using weights selected on other folds. The BCA aggregate describes performance over the 546 retained queries; it does not establish improvement across independent groups of questions linked by shared legal references.

Scores are reported to four decimal places, with scientific notation for small nonzero values. Differences are calculated before rounding. Each fine-tuning experiment uses its own pretrained baseline, and results from the two selection procedures are reported separately.

5.3 Experimental Results & Analysis

The results are organized into BCA dataset construction and retrieval, BGE-M3 fine-tuning, the overall retrieval benchmark, and a summary of the findings.

5.3.1 BCA Dataset Results

Collected Records and Target Coverage

The collection contains 652 distinct public question-answer records. Of these, 546 contain at least one extracted legal document reference and are retained for retrieval evaluation. The remaining 106 records are excluded because no reference was extracted. These excluded records is still a law question but the answers did not contain any legal references. The indexed corpus contains 14,789 articles.

Table 5.1: BCA collection size and target coverage.

Quantity	**Count**
Collected question-answer records	652
Retained evaluation queries	546
Indexed legal articles	14,789
Retained queries with all targets matched	211
Retained queries with some targets matched	229
Retained queries with no target matched	106

In total, 440 retained queries have at least one target matching an indexed article. Full coverage comprises 211 queries, calculated as 440 minus the 229 partially covered queries. Coverage measures identifier matches rather than mapping accuracy. No independently adjudicated mapping-accuracy estimate is available for this collection.

Observed Problems in Public-Service Questions

The collection records two distinct obstacles to constructing retrieval targets. Reference extraction yields no target for the 106 excluded records. These records is still a law question but the answers did not contain any legal references. This is a common problem in public-service questions, where the answer may describe a legal requirement without citing the relevant decree or article. The 229 partially covered queries illustrate another difficulty: the recorded targets are not fully matched to the indexed articles. the answers despite comming from official sources, still have incomplete or incorrect references. The 229 partially covered queries illustrate another difficulty: the recorded targets are not fully matched to the indexed articles.

These observations explain the difficulty of crawling and indexing public-service questions. The answers may not contain any legal references, or the references may be incomplete or incorrect.

Retrieval Benchmark on BCA

Table 5.2 compares BM25, pretrained BGE-M3, BM25+BGE-M3, and BM25+BGE-M3+E5 on all 546 retained BCA queries over a corpus of 14,789 legal articles. The results include questions with partial or no target coverage and summarize performance over the full retained collection. The grouping and selection procedure is described in Section 4.8.

Table 5.2: Retrieval results on 546 BCA queries over 14,789 legal articles.

Method	**Hit@10**	**MRR@10**	**MAP@10**	**NDCG@10**
BM25	0.4414	0.2434	0.1130	0.1687
BGE-M3	0.5220	0.3058	0.1469	0.2137
BM25+BGE-M3	0.5330	**0.3358**	**0.1606**	0.2283
BM25+BGE-M3+E5	**0.5495**	0.3348	0.1589	**0.2297**

Model BGE-M3 scores higher than BM25 on all four metrics, and both hybrids score higher than either individual retriever reported in the table. BM25+BGE-M3+E5 has the highest NDCG@10, the primary metric, and the highest Hit@10. BM25+BGE-M3 has the highest MRR@10 and MAP@10. Adding E5 therefore increases the observed NDCG@10 and Hit@10 but reduces the other two metrics; it does not improve every aspect of the ranking.

From this result we can say that hybrid retrieval improves the overall retrieval performance on the BCA dataset, with the combination of BM25, BGE-M3, and E5 yielding the best NDCG@10 and Hit@10 scores.

Another conclusion is, due to the nature of the BCA dataset, which contains public-service questions with everyday language, the result is sub 60% NDCG@10, which is lower than the other datasets. This indicates that the BCA dataset is more challenging for retrieval models, likely due to the informal language and incomplete or incorrect references in the questions and answers.

5.3.2 BGE-M3 Fine-Tuning Results

This section reports two fine-tuning experiments. The first selects BGE-M3 using hybrid retrieval performance. The second selects it using BGE-M3's standalone dense retrieval performance, then evaluates the selected model in the hybrids. The goal is to determine fine-tuning improves BGE-M3's retrieval performance, either as a standalone dense retriever or in hybrid configurations.

Selected Fine-Tuning Configurations

Table 5.3 lists the selected settings. The experiment run on the same dataset and iterate thought parameters to find the best configuration. The selected settings are based on the best performance of NDCG@10 on the development set. The selected settings are used for the final evaluation on the test set.

Table 5.3: Selected BGE-M3 fine-tuning parameters.

Parameter	**Hybrid-based selection**	**Standalone dense selection**
Training data	Zalo	Zalo
Learning rate	$\mathbf{2 \times 10^{-5}}$	$\mathbf{2 \times 10^{-5}}$
Training duration	**2 epochs**	**60 optimizer steps**
Contrastive query batch size	**4**	**64**
Effective optimizer batch	**32**	**64**
Explicit negatives / group	**7**	**5**
Temperature	**0.02**	**0.05**
Query / passage max length	**256 / 512 tokens**	**256 / 512 tokens**
Optimizer	AdamW	AdamW
Learning-rate scheduler	Linear	Cosine
Warmup ratio	**0.10**	**0.05**
Precision	FP16	BF16

Fine-Tuning for Hybrid Retrieval

Selecting the settings. Figure 5.1 shows development NDCG@10 for three learning rates at epochs 1, 2, and 3. The two panels report BM25+BGE-M3 and BM25+BGE-M3+E5 using the same metric scale.

Table 5.4: Query-batch comparison on Zalo development data.

Batch / steps	**BM25+BGE-M3**	**Δ**	**BM25+BGE-M3+E5**	**Δ**
Pretrained baseline	0.7172	N/A	0.7245	N/A
1 / 32	0.7725	+0.0553	0.7869	+0.0624
2 / 16	0.7693	+0.0521	0.7924	+0.0679
4 / 8	**0.7781**	**+0.0609**	**0.7934**	**+0.0689**

All three query-batch configurations increase development NDCG@10 for both hybrids relative to their pretrained baselines. The selected four-query batch has the highest score for each hybrid among these configurations.

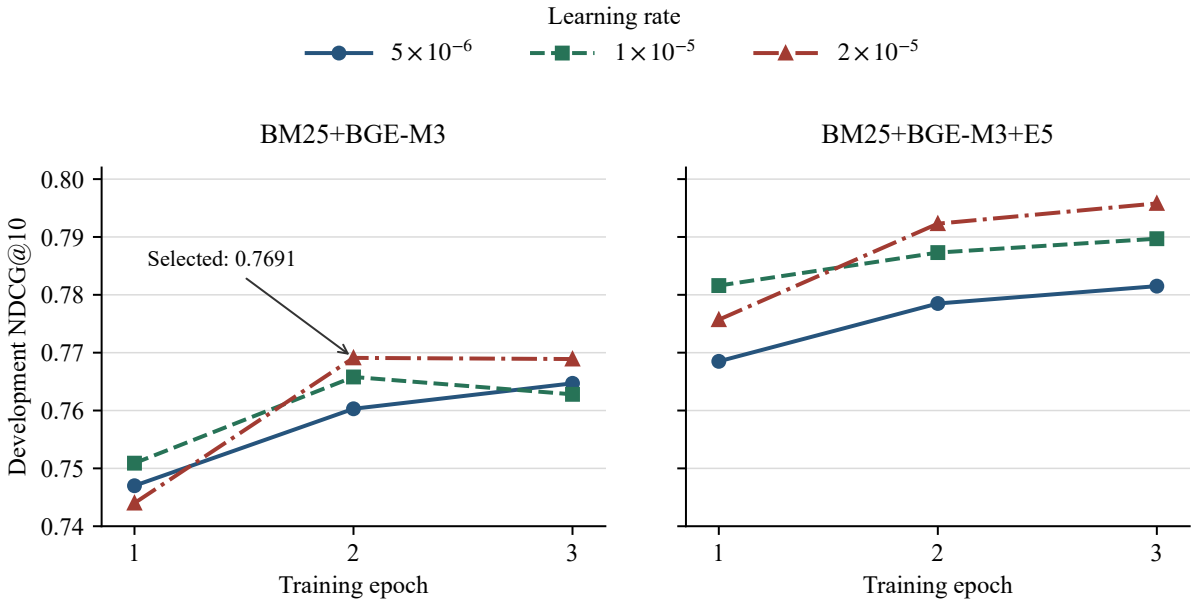


Figure 5.1: Development NDCG@10 by learning rate and training epoch on Zalo.

Table 5.5: BM25+BGE-M3 before and after fine-tuning on Zalo fold 0.

BGE-M3 state	**Seed**	**NDCG**	**Δ**	**MRR**	**MAP**	**Hit**
Pretrained	Fixed	**0.7516**	N/A	**0.7015**	**0.6985**	**0.9141**
Fine-tuned	42	0.7005	**−0.0511**	0.6470	0.6451	0.8750
Fine-tuned	43	0.7005	**−0.0511**	0.6490	0.6467	0.8688
Fine-tuned	44	0.6977	−0.0539	0.6398	0.6374	0.8859
Fine-tuned	Mean	0.6995	−0.0520	0.6453	0.6430	0.8766

All three fine-tuned BM25+BGE-M3 runs have lower NDCG@10 than their matched pretrained baseline. Their mean MRR@10, MAP@10, and Hit@10 are also lower.

Table 5.6: BM25+BGE-M3+E5 before and after fine-tuning on Zalo fold 0.

BGE-M3 state	**Seed**	**NDCG**	**Δ**	**MRR**	**MAP**	**Hit**
Pretrained	Fixed	**0.7595**	N/A	**0.7077**	**0.7048**	0.9266
Fine-tuned	42	0.7320	−0.0275	0.6771	0.6743	0.9141
Fine-tuned	43	0.7340	−0.0255	0.6771	0.6747	0.9203
Fine-tuned	44	0.7378	**−0.0217**	0.6794	0.6771	**0.9281**
Fine-tuned	Mean	0.7346	−0.0249	0.6779	0.6754	0.9208

BM25+BGE-M3+E5 also has lower NDCG@10 in all three fine-tuned runs. Its mean decrease is smaller than that of BM25+BGE-M3. Seed 44 has higher Hit@10 than the pretrained baseline but lower NDCG@10 and MRR@10.

Fine-Tuning for Standalone Dense Retrieval

Selecting the settings. Selection evaluates 36 checkpoints across three learning rates, four development folds, and steps 30, 60, and 90, following Section 4.7.2. The selected configuration uses Zalo training data, learning rate 2×10^{-5}, and 60 optimizer steps.

Table 5.7 reports the selected development scores. NDCG@10 improves in all four folds, and the configuration meets the selection criteria. Later data-expansion stages are therefore not run.

Table 5.7: Development results for the selected dense model.

Quantity	**Selected result**
Mean development NDCG@10	0.7504
NDCG@10 gain over the pretrained development baseline	+0.0605
Development MRR@10	0.7111
Development Recall@100	0.9633

Standalone test results. Table 5.8 compares pretrained and fine-tuned BGE-M3 on the official Zalo test dataset of 640 queries using extracted article text. This comparison uses a different test allocation and pretrained baseline from the hybrid-selected experiment. The test queries had been examined earlier, so the results are interpreted as post-hoc evidence.

Table 5.8: BGE-M3 before and after fine-tuning on the official Zalo test dataset.

System	**NDCG@10**	**Delta NDCG@10**	**MRR@10**	**Recall@100**
Pretrained	0.6740	0.0000	0.6213	**0.9617**
Fine-tuned BGE-M3	**0.6932**	**0.0192**	**0.6448**	0.9531

All three runs improve NDCG@10 and MRR@10 but reduce Recall@100. Mean NDCG@10 increases by 0.0189, while mean Recall@100 decreases by 0.0091. No statistical significance test is reported for this standalone comparison.

Hybrid test results. Table 5.9 evaluates the selected model in both hybrids using the same test queries and corpus. BM25, E5, and fusion weights remain fixed. The weights are 0.20/0.80 for BM25/BGE-M3 and 0.15/0.50/0.35 for BM25/BGE-M3/E5, selected using pretrained BGE-M3.

Relative Δ is the change divided by the pretrained score. Positive-seed counts record any NDCG@10 increase, regardless of the improvement threshold.

Table 5.9: Hybrid results after selection by standalone dense performance.

Quantity	**BM25+BGE-M3**	**BM25+BGE-M3+E5**
Pretrained NDCG@10	**0.7526**	0.7467
Fine-tuned mean NDCG@10	0.7483	**0.7476**
Δ NDCG@10	−0.0043	+0.0009
Relative Δ NDCG@10	−0.5707%	+0.1203%
Δ Recall@100	−0.0057	−0.0031
Seeds with positive Δ NDCG@10	0/3	2/3
NDCG@10 improvement threshold	Not met	Not met

Mean NDCG@10 decreases by 0.0043 for BM25+BGE-M3 and increases by 0.0009 for BM25+BGE-M3+E5. Neither reaches the predefined improvement threshold of 0.01, and Recall@100 decreases for both. Therefore, the standalone improvement does not produce a practically useful hybrid improvement under fixed fusion weights.

Changes in Individual Queries

Table 5.10 compares per-query NDCG@10 before and after hybrid-selected fine-tuning on Zalo dataset. Table 5.11 reports the standalone-selected experiment on the official Zalo test dataset, averaging each query's scores across seeds 42, 43, and 44. Each comparison uses 640 queries and its own pretrained baseline, with fusion weights fixed.

Queries are classified by whether NDCG@10 increases, decreases, or remains unchanged. A top-ten miss means that no recorded target is retrieved in the first ten results. The miss counts in Table 5.10 are reported separately from the NDCG@10 categories.

Table 5.10: Query-level changes after hybrid-based selection.

Outcome	**BM25+BGE-M3**	**BM25+BGE-M3+E5**
Higher NDCG@10	102	102
Lower NDCG@10	210	163
Unchanged NDCG@10	328	375
Previously missing at ten, now retrieved	23	16
Previously retrieved at ten, now missing	48	24
Missing at ten in both systems	32	31

After hybrid-selected fine-tuning, both hybrids have more queries with lower NDCG@10 than higher NDCG@10, and introduce more top-ten misses than they recover.

Table 5.11: Query-level changes after standalone dense selection.

Comparison	Improved	Worse	Tied
Dense	127	96	417
BM25+BGE-M3	81	93	466
BM25+BGE-M3+E5	85	86	469

After standalone-selected fine-tuning, dense retrieval improves more queries than it worsens. Both hybrids show the opposite pattern, with most queries unchanged.

Summary of Fine-Tuning Results

The hybrid-selected configuration increases development scores but lowers test NDCG@10 for both hybrids in every reported seed. The standalone-selected configuration increases dense NDCG@10 and MRR@10 in every seed but lowers Recall@100. Neither corresponding hybrid reaches the predefined +0.01 NDCG@10 improvement threshold.

These post-hoc results show an observed dense ranking gain without a practically useful hybrid gain under the fixed fusion weights. The experiments use different training procedures and test allocations, so their differences cannot be attributed to a single parameter. Retuned fusion weights and transfer to ALQAC or BCA were not evaluated.

5.3.3 Overall Retrieval Benchmark

This section compares the pretrained retrieval methods on the supplied ALQAC and Zalo reference corpora. The BCA results in Section 5.3.1 form the public-service comparison; the fine-tuning results in Section 5.3.2 remain separate because they use different selection procedures and matched test populations.

Benchmark Conditions

The overall benchmark uses only the reference article text supplied with ALQAC and Zalo: 811 queries and 3,363 articles for ALQAC, and 3,196 queries and 61,425 articles for Zalo. Within each dataset, all methods use the same questions, corpus, and recorded relevance targets. Each query receives equal weight and one out-of-fold prediction under the grouped selection procedure in Section 4.8.

Lexical and Hybrid Retrieval

Table 5.12 compares BM25, pretrained BGE-M3, pretrained multilingual-E5-base (E5), and the hybrid configurations on ALQAC and Zalo. Table 5.13 reports their NDCG@10 changes relative to BM25.

Table 5.12: Pretrained retrieval results on ALQAC and Zalo.

Method	**NDCG@10**	**MRR@10**	**MAP@10**	**Hit@10**
ALQAC (811 queries)				
BM25	0.7986	0.7648	0.7621	0.9112
BGE-M3 pretrained	0.8437	0.8134	0.8108	0.9433
E5 pretrained	0.8126	0.7729	0.7722	0.9371
BM25+BGE-M3	**0.8805**	**0.8534**	**0.8516**	**0.9692**
BM25+BGE-M3+E5	0.8784	0.8519	0.8504	0.9630
Zalo (3,196 queries)				
BM25	0.5264	0.4731	0.4682	0.7100
BGE-M3 pretrained	0.6424	0.5917	0.5850	0.8235
E5 pretrained	0.5640	0.5104	0.5046	0.7509
BM25+BGE-M3	0.6805	0.6298	0.6235	**0.8583**
BM25+BGE-M3+E5	**0.6818**	**0.6317**	**0.6254**	0.8576

BGE-M3 has the highest NDCG@10 among the individual retrievers on both datasets. Both hybrids score higher than BM25 and the individual dense retrievers.

Table 5.13 reports absolute NDCG@10 changes as hybrid minus BM25. Relative changes divide these differences by the corresponding BM25 NDCG@10.

Table 5.13: Hybrid NDCG@10 gains over BM25.

Dataset	**Hybrid**	**Absolute change**	**Relative change**
ALQAC	BM25+BGE-M3	**+0.0819**	**+10.2590%**
ALQAC	BM25+BGE-M3+E5	+0.0798	+9.9970%
Zalo	BM25+BGE-M3	+0.1541	+29.2760%
Zalo	BM25+BGE-M3+E5	**+0.1554**	**+29.5210%**

The absolute NDCG@10 gains over BM25 are larger in Zalo than in ALQAC.

Contribution of E5

The contribution of E5 is measured by comparing BM25+BGE-M3+E5 with BM25+BGE-M3 on the same evaluation queries. Absolute changes subtract the BM25+BGE-M3 NDCG@10 value; relative changes divide by that same baseline.

Table 5.14: NDCG@10 changes from adding E5.

Dataset	Absolute change	Relative change
ALQAC	-0.0021	-0.2380%
Zalo	+0.0013	+0.1900%

Table 5.14 shows a small NDCG@10 decrease on ALQAC and a small increase on Zalo. The observed contribution of E5 therefore differs between the two datasets.

Highest-Scoring Retrieval Configurations

On ALQAC, BM25+BGE-M3 has the highest observed NDCG@10, MRR@10, MAP@10, and Hit@10. Adding E5 slightly reduces all four metrics.

On Zalo, BM25+BGE-M3+E5 has the highest NDCG@10, MRR@10, and MAP@10, while BM25+BGE-M3 has the highest Hit@10.

For the full 546-query BCA collection, BM25+BGE-M3+E5 has the highest observed NDCG@10 and Hit@10, whereas BM25+BGE-M3 has the highest MRR@10 and MAP@10. These descriptive results do not establish an advantage across independent legal groups.

Across the evaluated conditions, both hybrids have higher NDCG@10 than BM25. The contribution of E5 varies by condition and metric. Overall, the highest-scoring configuration depends on the metric and dataset. These results do not establish a general E5 improvement.

5.3.4 Summary of Results

From the overall benchmark, pretrained BGE-M3 has higher NDCG@10 than BM25 and E5 on both ALQAC and Zalo.

Both hybrids BM25+BGE-M3 and BM25+BGE-M3+E5 have higher NDCG@10 than BM25 and the individual dense retrievers. The contribution of E5 is small and varies by condition.

The finetune version of BGE-M3 does not improve hybrid retrieval in both dataset experiments. This provides evidence that the pretrained BGE-M3 is already well-suited for hybrid retrieval in the legal domain, and further fine-tuning does not yield significant improvements.

Chapter 6

Conclusion and Future Work

This chapter synthesizes the experimental findings rather than repeating the numerical analysis from Chapter 5. The conclusions distinguish improvement over BM25, incremental value beyond an existing hybrid, and transfer from dense adaptation into a complete retrieval system.

6.1 Summary of Findings

6.1.1 BCA Dataset Construction and Coverage

The BCA collection contains 652 public question-answer records, of which 546 have extracted legal references and are retained for evaluation against 14,789 indexed articles. Among the retained queries, 211 have all targets available, 229 have some targets available, and 106 have none available (Table 5.1).

6.1.2 Hybrid Retrieval Compared with BM25

Both BM25+BGE-M3 and BM25+BGE-M3+E5 have higher NDCG@10 than BM25 on the ALQAC and Zalo datasets and the full 546-query BCA collection (Tables 5.12 and 5.2). BGE-M3 also has higher NDCG@10 than BM25 as an individual retriever on all three collections. The observed hybrid gains support combining lexical and dense scores will most likely improve retrieval performance.

6.1.3 Contribution of E5

Adding E5 to BM25+BGE-M3 changes NDCG@10 by approximately -0.0021 on ALQAC and +0.0013 on Zalo. Neither reported query-level test provides sufficient evidence of an improvement (Table 5.14). For the full BCA collection, NDCG@10 increases from 0.2283 to 0.2297 and Hit@10 increases, while MRR@10 and MAP@10 decrease (Table 5.2). These mixed results suggest that E5 adding second to none improment to the hybrid system with BM25+BGE-M3.

6.1.4 BGE-M3 Fine-Tuning

The thesis evaluates two ways of selecting a fine-tuned BGE-M3 model: using hybrid retrieval performance or using BGE-M3's standalone retrieval performance. In both experiments, the hybrid comparison replaces pretrained BGE-M3 with its fine-tuned version while keeping BM25, E5, and the combination weights fixed.

When model selection uses hybrid performance, fine-tuning improves development scores but gives lower mean NDCG@10 on the test dataset. The decreases are 0.0520 for BM25+BGE-M3 and 0.0249 for BM25+BGE-M3+E5 (Tables 5.5 and 5.6). These are observed changes on the test dataset; the reported uncertainty intervals include zero and do not establish a general decrease in hybrid performance (Table **??**).

When model selection uses standalone performance, fine-tuning increases BGE-M3's mean NDCG@10 by 0.0189, although Recall@100 decreases by 0.0091 (Table 5.8). The hybrid results show a decrease of 0.0043 for BM25+BGE-M3 and an increase of 0.0009 for BM25+BGE-M3+E5. Neither meets the predefined NDCG@10 improvement threshold of 0.01 (Table 5.9).

Therefore, the thesis concludes that BGE-ME can be finetuned to improve its own retrieval performance on Vietnamese legal data,but not necessarily in the hybrid setting.

6.2 Research Contributions

The first contribution is an empirical evaluation of hybrid retrieval for Vietnamese legal articles on the ALQAC, Zalo, and BCA datasets. The thesis compares BM25+BGE-M3 and BM25+BGE-M3+E5 with BM25 using the same queries, indexed articles, and relevance targets within each evaluation condition. Comparing the two hybrids separately identifies the observed contribution of E5. (Sections 5.3.1 and 5.3.3).

The second contribution is an evaluation of fine-tuning the BGE-M3 embedding model for standalone and hybrid retrieval. Two experiments select the model using hybrid retrieval performance and standalone dense retrieval performance on development data. The hybrid comparisons keep BM25, E5, and the combination weights fixed when replacing pretrained BGE-M3 with its fine-tuned version. This evaluation tests whether fine-tuning improves BGE-M3 on its own and whether the selected model improves the complete hybrid system (Section 5.3.2).

The third contribution is the construction and evaluation of the BCA dataset from public-service questions and published answers. The thesis documents the collection, cleaning, reference extraction, and identifier mapping procedures used to turn these records into retrieval queries and relevance targets (Section 4.3). Its evaluation examines target availability in the corpus and accounts for links between questions that share legal

references. This provides a public-service dataset alongside the competition benchmarks and a basis for distinguishing corpus-coverage and mapping problems from ranking errors (Section 5.3.1).

6.3 Discussion and Limitations

The observed gains over BM25 support hybrid retrieval on the evaluated datasets. However, the contribution of E5 varies by dataset and metric, so a higher score than BM25 is not sufficient evidence for adding another embedding model. The relevant comparison is BM25+BGE-M3+E5 against BM25+BGE-M3 under the same evaluation conditions. The reported results do not establish one hybrid configuration as the best choice for every legal dataset.

The fine-tuning comparisons evaluate replacement of pretrained BGE-M3 while keeping BM25, E5, and the combination weights fixed. They therefore show how the selected encoder performs within an existing hybrid configuration. They do not determine whether selecting new weights after fine-tuning would improve the results. The two experiments also use different training procedures and model-selection criteria, so their outcomes cannot be attributed to a single training parameter. Neither retuned hybrid weights nor transfer of the fine-tuned models to ALQAC or BCA was evaluated (Section 5.3.2).

BCA introduces limitations in both target construction and corpus coverage. Its relevance targets are extracted from published answers, which may omit supporting articles or contain references that are extracted or mapped incorrectly. An article absent from the corpus cannot be retrieved, while an incorrect identifier mapping can prevent a retrieved article from matching its target. Queries with partial coverage retain targets that are unavailable. Consequently, the reported metrics reflect corpus availability and target matching as well as ranking performance (Section 5.3.1).

BCA also show the difficulty of applying retrieval to answer public-service questions. Many queries have no available targets, and some have only a subset of their targets in the corpus. The questions are often long, complex, and contain multiple sub-questions. The answers are often long and contain multiple references. These characteristics make the queries more difficult than the competition benchmarks, which were designed to be answerable by a single article.

Both fine-tuning test datasets had been examined in earlier analyses. Although the corresponding test queries are excluded from training and model selection, the results remain post-hoc evidence. Repeating training with three seeds measures variation between the observed training runs on the same test dataset; it does not establish that the same changes would occur on a new set of questions.

The evaluation is limited to Vietnamese legal article retrieval using the recorded relevance

targets. It does not assess retrieval in other languages or legal systems. It also does not measure generated-answer correctness, the current applicability of legal provisions, user satisfaction, or deployment latency, throughput, and cost. These outcomes require separate evaluation before the findings can support claims about a complete legal question-answering service.

6.4 Future Work

A new, prospectively controlled evaluation should be conducted on an untouched query set. Corpus construction, model selection, fusion configuration, and statistical procedures should be fixed before examining the final test results.

Corpus quality should also be improved, particularly for the BCA collection. Coverage gaps can be addressed through version-aware mapping of legal documents, while query-level failures can be systematically sampled and annotated using a defined rubric. This would help distinguish missing evidence, annotation limitations, lexical mismatch, semantic confusion, and fusion-related ranking errors.

Future work should further extend the evaluation from retrieval performance to the complete RAG system. This includes assessing answer correctness, citation support, legal sufficiency, latency, throughput, and computational cost. Retrieval metrics remain useful indicators, but they cannot replace direct evaluation of generated legal answers and deployed system performance.

6.5 Closing Remarks

This thesis shows that Vietnamese legal retrieval benefits from lexical–semantic hybridization, but also that more components or better dense-only scores do not automatically produce a better complete system. The most important lesson is methodological: legal retrieval claims must be tied to the exact corpus, target coverage, selection boundary, and system level being evaluated. A disciplined negative result can be as informative as a positive result when it prevents an unsupported deployment claim.

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